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Subject : PWR Benchmark

and a critical boron concentration dependent on CA configuration

(see attached Table)

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NEACRP-L-335: 3-D LWR Core Transient Benchmark Specification

Addendum to PWR Benchmark Specification

Table 2.5 is split up into Tables 2.5.1 and 2.5.2

.373220E-02	-.319253E-02	.247770E-02	-.102786E-03	-.377989E-04	
-.219926E-01	.255875E-01	-.282319E-02	-.115483E-02	absorber type1	
.374092E-02	-.314239E-02	.242926E-02	-.122634E-03	-.459250E-04	
-.167503E-01	.256478E-01	-.328086E-02	-.134262E-02	absorber type2	

Table 2.5.1 – Cross sections $\Delta\Sigma_{CA}$ of control assemblies in axial layers 2 through 17 (active core)

.373220E-02	-.319253E-02	.247770E-02	.000000E+00	.000000E+00	
-.219926E-01	.255875E-01	.000000E+00	.000000E+00	absorber type1	
.374092E-02	-.314239E-02	.242926E-02	.000000E+00	.000000E+00	
-.167503E-01	.256478E-01	.000000E+00	.000000E+00	absorber type2	
.697102E-02	-.119034E-02	.879034E-04	.000000E+00	.000000E+00	
-.113498E-01	.170043E-02	.000000E+00	.000000E+00	driver device	

Table 2.5.2 – Cross sections $\Delta\Sigma_{CA}$ of control assemblies in axial layer 18 (upper axial reflector)

NEACRP 3-D LWR CORE TRANSIENT BENCHMARK
Final Specifications

by

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and

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Rome, Italy

October 1991 (January 1992)

OECD Nuclear Energy Agency

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TABLE OF CONTENTS

	Page
1. Introduction	3
2. Reference Pressurized Water Reactor	5
2.0. General	5
2.1. Core Geometry	5
2.2. Neutron Modeling	6
2.3. Macroscopic Cross Sections and Derivatives	6
2.4. Composition Map	7
2.5. Doppler Temperature	7
2.6. Fuel Assembly Geometry	7
2.7. Thermophysical Properties	8
2.8. Neutronics-Thermohydraulics Coupling	8
2.9. Operation Data	8
2.10. Heat Exchange Correlations	9
2.11. Pressure Drops	9
3. PWR Problems	10
3.0. Nature of the Problems	10
3.1. Calculations of the Initial Steady State	10
3.2. Transient Calculations	10
4. PWR Problems: Output Requested	12
5. Reference Boiling Water Reactor	13
5.0. General	13
5.1. Core Geometry	13
5.2. Neutron Modeling	13
5.3. Macroscopic Cross Sections and Derivatives	14

94310005

	Page
5.4. Composition Map	15
5.5. Doppler Temperature	15
5.6. Macroelement Geometry	16
5.7. Thermophysical Properties	16
5.8. Neutronics-Thermohydraulics Coupling	16
5.9. Operation Data	16
5.10. Heat Exchange Correlations	16
5.11. Pressure Drops	17
6. BWR Problems	18
6.0. Nature of the Problems	18
6.1. Calculations of the Initial Steady State	18
6.2. Transient Calculations	18
7. BWR Problems: Output Requested	20
7.1. Steady State Results	20
7.2. Transient Results	20
8. Common Remarks for Output Format	21
List of Tables	22
List of Figures	24

1. Introduction

Mathematical benchmarks, based on well defined problems with a complete set of input data and a unique solution, are widely used and accepted means of verifying the reliability of numerical simulations, i.e. the accuracy, stability and efficiency of numerical methods. Problems are often very testing, but tend to be somewhat simplified - in order to make the analysis manageable - when the purpose is the intercomparison of several different models.

This is the case for the present benchmark, which is aimed at assessing the discrepancies between three-dimensional codes for transient calculations in Light Water Reactor cores. Thanks to the development of advanced numerical methods for larger and more efficient computers and to the increased interest in accurate core dynamics simulations, quite a few such codes are now available in various OECD countries. So, NEACRP's initiative to promote this international benchmark, appears timely, and likely to attract a large participation.

The reference problem chosen for simulation in a PWR is the ejection of a control assembly from the core, which may occur as a consequence of the rupture of the drive mechanism casing located on the reactor pressure vessel top. This event can produce significant, well localized perturbations of the neutronic and thermohydraulic core parameters, without exceeding the safety margins. Hence, a rather realistic standard reactor situation is defined, that efficiently utilizes the neutronics and thermohydraulics submodels of the reactor dynamics code.

The BWR problems consist in sub-promptcritical reactivity excursions generated by rapid cold water injection or core pressurization events. This set of problems was chosen because it is felt that the analysis of two such cases - in which the interplays of the neutronic and thermohydraulic effects are markedly different - represents a direct and effective way to fulfil the objectives of the benchmark.

Most problems in this benchmark are also suitable for one-dimensional calculations. It is strongly recommended to participants to supply 1-D solutions (using their own

condensation "recipes") together with the reference 3-D solutions. An extensive assessment of the accuracies of 1-D against 3-D approximation schemes could prove a very useful exercise.

In the following sections the complete set of input data is given for both PWR and BWR problems. This must be considered a draft, as the "final" data will be provided on request to participants in diskette form.

2. Reference Pressurized Water Reactor

2.0. General

The reference scheme for the Pressurized Water Reactor (PWR) is derived from real reactor geometry and operation data. One modification was introduced, consisting in the addition of a central control assembly (CA), which allowed us to set up the problem of a single rod ejection from a core octant with full rotational symmetry.

The set of data given in the following paragraphs and in the pertinent tables and figures, completely defines the three pairs of PWR benchmark exercises.

2.1. Core Geometry

The radial geometry of the reactor core is shown in Figure 2.1. Radially, the core is divided into cells of 21.606 cm width, each corresponding to one fuel assembly (FA), plus a radial reflector cell (shaded area) of the same width. Axially, the reactor core is divided into 16 layers with heights of 7.7, 11.0, 15.0, 30.0 (10 layers), 12.8 (2 layers) and 8.0 cm (from bottom to top), adding up to a total height of the active core of 367.3 cm. Upper and lower axial reflector have thicknesses of 30.0 cm.

Fuel assemblies with different U-235 enrichments and different numbers of rods of burnable absorbers are present in the core. The axial and radial distributions of the enrichments and absorbers can be found in chapter 2.4.

The radial arrangement of control assemblies is shown in Figure 2.2. The total CA length, which coincides with the absorber length, is 362.159 cm. The driver device section following the top of the absorbers is distinguished from the absorber via a different cross section data set. No tip of control rods is defined. The position of the lower CA absorber edge from the bottom of the lower reflectors is 37.7 cm for a completely inserted CA, and 401.183 cm for a completely withdrawn CA. Measured in units of steps, complete insertion or withdrawal of a CA corresponds to 0 and 228 steps, respectively.

2.2. Neutron Modeling

Two prompt neutron groups, i.e. thermalized and fast neutrons, and six delayed neutron groups are used for neutron modeling. The boundary condition for the solution of the neutron diffusion equation is flux vanishing at the outer reflector surface.

Velocities and the energy release per fission for the two prompt neutron groups are given in Table 2.1, and are considered to be independent in time and space. Table 2.2 shows the time constants and fractions of delayed neutrons. No delayed energy release is considered.

2.3. Macroscopic Cross Sections and Derivatives

A complete set of macroscopic cross sections for transport, scattering, absorption and fission and their derivatives with respect to boron density, moderator temperature, moderator density, and fuel temperature is defined for each composition. Table 2.4 shows the definition of all cross sections, derivatives, and reference values associated with a composition.

Let ρ be the water density, T_F the Doppler temperature, c the boron concentration in ppm, and T_M the moderator temperature. Any macroscopic cross section Σ must be calculated according to the formula

$$\begin{aligned}\Sigma = & \Sigma_o + (\partial\Sigma/\partial\rho)_o (\rho - \rho_o) + (\partial\Sigma/\partial\sqrt{T_F})_o (\sqrt{T_F} - \sqrt{T_{Fo}}) + \\ & + (\partial\Sigma/\partial c)_o (c - c_o) + (\partial\Sigma/\partial T_M)_o (T_M - T_{Mo})\end{aligned}$$

where subscript "o" denotes the reference values ρ_o , T_{Fo} , c_o and T_{Mo} and the reference point $(\rho_o, T_{Fo}, c_o, T_{Mo})$.

The cross section at a numerical node with CA is determined by adding the cross section $\Delta\Sigma_{CA}$ contributed from the CA to the cross section without CA:

$$\Sigma_{\text{with CA}} = \Sigma_{\text{without CA}} + P \Delta\Sigma_{CA}$$

where p is the relative insertion in the node, i.e. $0 \leq p \leq 1$. The contribution of the CA driver is treated in an analogous way. The incremental cross section for the CAs (absorber type 1: in 2.1 % enriched FAs; absorber type 2: in 3.1 % FAs) and for their drivers, are given in Table 2.5 with the same key as in Table 2.4.

2.4. Composition Map

Within the core geometry 11 different compositions and the corresponding sets of cross sections are defined. The definitions of the compositions can be seen in Table 2.3. Each space cell of the reactor geometry can be related to one of these compositions. The axial and radial locations of the compositions within the reactor geometry are shown in Figures 2.3 through 2.5. The values of cross sections and derivatives which are associated with each composition are defined in Tables 2.6.1 through 2.6.11. The key to these tables is explained in Table 2.4.

2.5. Doppler Temperature

The Doppler temperature T_D is found from the fuel temperature at the fuel rod center $T_{F,C}$ and on the fuel rod surface $T_{F,S}$ via the relation

$$T_D = (1-\alpha) T_{F,C} + \alpha T_{F,S}$$

where α is taken equal 0.7.

2.6. Fuel Assembly Geometry

The geometrical data for the FA are given in Table 2.7.

2.7. Thermophysical Properties

The UO_2 density without dishing is 10.412 g/cm^3 (95 % of the theoretical density) at a temperature of 20°C . The pellet dishing amounts to 1.248 %. The cladding material is Zircaloy-4 with a density of 6.6 g/cm^3 .

Participants of the benchmark should use the following reference relations for the heat conductivity λ ($\text{W/m}^\circ\text{K}$) and specific heat capacity c_p ($\text{J/Kg }^\circ\text{K}$) of fuel and cladding

$$\lambda_{\text{UO}_2} = 1.05 + 2150/(T-73.15)$$

$$\lambda_{\text{Zircaloy-4}} = 7.51 + 2.09 \cdot 10^{-2} T - 1.45 \cdot 10^{-5} T^2 + 7.67 \cdot 10^{-9} T^3$$

$$c_{p,\text{UO}_2} = 162.3 + 0.3038 T - 2.391 \cdot 10^{-4} T^2 + 6.404 \cdot 10^{-8} T^3$$

$$c_{p,\text{Zircaloy-4}} = 252.54 + 0.11474 T$$

where T is the temperature ($^\circ\text{K}$).

Expansion effects of fuel and cladding will not be considered in this benchmark.

2.8. Neutronics-Thermohydraulics Coupling

The feedback or coupling between neutronics and thermohydraulics is characterized by the definition of channel regions. In the present work it is recommended to define each FA as a channel region.

A flat profile of the radial distribution of the power density inside the fuel shall be assumed.

2.9. Operation Data

The reactor is at the beginning of cycle 1 (zero EPFD: no Xenon or Iodine, no fuel depletion). The steady state operation data are defined in Table 2.8. The thermal energy

output is to be released for 98.1 % in the fuel and for 1.9 % in the coolant.

The inlet mass flow through the core given in Table 2.8 is distributed uniformly among the channels.

2.10. Heat Exchange Correlations

The conductance of the helium-filled gap between fuel and cladding (k_{gap}) is assumed to be constant:

$$k_{\text{gap}} = 10^4 \text{ W/m}^2 \text{ } ^\circ\text{K}$$

2.11. Pressure Drops

A homogeneous core pressure of 155 bar is assumed.

3. PWR Problems

3.0. Nature of the Problems

The transient to be analyzed as a function of time in three space dimensions are generated by the rapid ejections of a control assembly (CA) from an initially delayed critical core at hot zero power (HZP) or at full power (FP). The set of realistic problems offers a variety of reactivity excursions -from about 0.1\$ to about 1.1\$- that are expected to efficiently test both the thermal-hydraulic and neutronic models of reactor dynamics codes. With respect to axially one-dimensional solutions, which are also of great interest in this benchmark, the cases are meant to present increasing levels of difficulty for such approximation.

3.1. Calculations of the Initial Steady State

In order to achieve an effective multiplication factor of one, the critical steady state parameters of the reactor core have to be found from a search calculation of the critical boron concentration for the given thermal power and CA configuration, and for the parameters defined in Tables 2.7 and 2.8.

3.2. Transient Calculations

Six cases (or, better, three pairs of cases) are submitted for the benchmark calculations. They are as follows:

- Case A1 : (Figure 3.1). Core octant with rotational symmetry. Ejection of a central CA (circled) at HZP.
- Case A2 : (Figure 3.2). Same as above at FP.
- Case B1 : (Figure 3.3). Core octant with rotational symmetry. Ejection of a peripheral CA (circled) at HZP.

Case B2 : (Figure 3.4). Same as above at FP.

Case C1 : (Figure 3.5). Full core. Ejection of a peripheral CA (circled) at HZP.

Case C2 : (Figure 3.6). Same as above at FP.

The time for CA ejection is 100 ms for all cases independent of the initial insertion depth, thus causing a transient with a time scale of a few seconds. After ejection has occurred no reactor scram will be considered. During the whole calculation, the boron concentration and the positions of the (other) CAs are kept constant. The boron concentration in each case is to be selected as the critical boron concentration of the respective critical initial state.

A drawing of the initial CA configuration and drawing of CAs which are to be ejected for each case are shown in Figures 3.1 through 3.6. The CA to be ejected is marked by circle. The CAs are labeled with symbols —, C, B, A and X which correspond to insertion lengths of 228, 200, 150, 100 and 0 steps. Those configurations will meet the requirement for the reactivity change as mentioned above.

4. PWR Problems: Output Requested

Results should be presented on paper and diskette in the following form (some more details are given in Section 8):

A) specification of case (case A1 - case C2)

B) "steady state results"

B1) "critical boron concentration:"

B2) "radial power distribution:" (normalized to the peak value equal to 1)

B2-1) "at axial layer number 6:" (of the active core: see Figures 8.1-8.3)

B2-2) "at axial layer number 13:"(of the active core: see Figures 8.1-8.3)

B3) "maximum power peaking factor:"

B4) "position of maximum power peaking factor:"

B5) "axial power distribution:" (core averaged; normalized to the peak value equal to 1: see Figure 8.1)

C) "transient results"

C1) "core power versus time:" (normalized to the steady state power; 0-5 s)

C2) "core averaged fuel temperature versus time:" (0-5 s)

C3) "maximum fuel temperature versus time:" (0-5 s)

C4) "coolant outlet temperature versus time:" (core averaged; 0-5 s)

C5) "radial distribution of power at time of power maximum:"

(normalized to the peak value equal to 1.)

C5-1) "at axial layer number 6:" (of the active core: see Figures 8.1-8.3)

C5-2) "at axial layer number 13:"(of the active core: see Figures 8.1-8.3)

C6) "radial distribution of power at final time $t = 5$ s:"

(normalized to the peak value equal to 1.)

C6-1) "at axial layer number 6:" (of the active core: see Figures 8.1-8.3)

C6-2) "at axial layer number 13:"(of the active core: see Figures 8.1-8.3)

5. Reference Boiling Water Reactor

5.0. General

The reference scheme for the BWR problems is directly derived from existing reactors. Some minor changes have been introduced, with the purpose of making life easier for most 3-D core dynamics codes. This is the case for the definition of a macroelement -a homogeneous average of four real BWR fuel elements with the pertinent control rod in the middle- which simplifies the core configuration and reduces the minimum number of nodes in the X-Y plane by a factor of four.

5.1. Core Geometry

The X-Y reactor geometry is shown in Figure 5.1. The side of the square fuel macroelement, as well as of the analogous macrocell in the radial reflector, is 30.48 cm. Axially, the reactor is subdivided into 14 layers, each 30.48 cm high, as shown in Figures 5.2 through 5.10. The largest acceptable mesh in the solution schemes will therefore be 30.48 x 30.48 x 30.48 cm.

5.2. Neutron Modeling

Two prompt neutron energy groups and six delayed neutron groups are used for neutron modeling.

Table 5.1 gives the mean prompt neutron inverse velocities v_1^{-1} and v_2^{-1} and the prompt energy release per fission, E_0 . These values are to be considered space and time independent. No delayed energy release is considered.

Table 5.2 gives the delayed neutron fractions β_i and the time constants λ_i ($i=1,\dots,6$).

The neutron fluxes are assumed to vanish on the reactor boundaries.

5.3. Macroscopic Cross Sections and Derivatives

Let ρ and T be the water density and the Doppler temperature respectively. Any macroscopic cross section Σ (transport, absorption, fission, scattering) must be calculated following the formula

$$\Sigma = \Sigma_0 + \Sigma_\rho (\rho - \rho_0) + \Sigma_T (\sqrt{T} - \sqrt{T_0})$$

where:

- ρ_0 and T_0 are reference values of water density and Doppler temperature respectively;

- Σ_0 is the value of Σ at the reference point $P_0 \equiv (\rho_0, T_0)$;

- Σ_ρ is the derivative of Σ with respect to water density, at the reference point $P_0 \equiv (\rho_0, T_0)$;

- Σ_T is the derivative of Σ with respect to the square root of the Doppler temperature, at the reference point $P_0 \equiv (\rho_0, T_0)$.

The above equation leads to the definition not only of a nuclear composition as a complete set of two-group macroscopic cross sections ($\Sigma_{t,1}, \Sigma_{1 \rightarrow 2}, \Sigma_{a,1}, v\Sigma_{f,1}, \Sigma_{f,1}, \Sigma_{t,2}, \Sigma_{a,2}, v\Sigma_{f,2}, \Sigma_{f,2}$) at the reference point P_0 , but also of a generalized nuclear composition as a complete set of macroscopic cross sections and of their derivatives at the reference point P_0 .

Table 5.3 shows the list of data characterizing a generalized nuclear composition, including the reference values ρ_0 and T_0 and an integer to identify the composition. Table 5.3 also shows the key to Tables 5.4 through 5.22.

Tables 5.4 through 5.22 give the data of the generalized nuclear compositions included in our BWR problems. These data take into account all the materials that are

present in the core (fuel, coolant, structures, control and burnable absorbers).

5.4. Composition Map

To relate the core geometry with the generalized nuclear compositions, 10 types of macroelements (including the radial reflector macroelement) were distinguished in the map of Figure 5.1. For each macroelement type, a number (Composition Identifier) is associated to each layer in Figures 5.2 through 5.10. Number 19 is associated to all radial reflector layers.

So, the 3-D nuclear composition map is completely defined, in the sense that a Composition Identifier is associated to all 3-D meshes of our reactor. That allows the participants to calculate the actual macroscopic cross sections in all spatial meshes both in the steady state and in the transient calculations, by knowing the local water density and Doppler temperature.

Obviously, the macroscopic cross sections and their derivatives, as given in the generalized nuclear compositions, are homogenized over the mesh volume, so that the internal structure of a macroelement is useless from the neutronics point of view.

5.5. Doppler Temperature

The Doppler temperature to be used in the macroscopic cross section calculations is related to the actual temperature by the formula

$$T = (1-\alpha) T_{F,C} + \alpha T_{F,S}$$

where $T_{F,C}$ and $T_{F,S}$ are the fuel temperatures at the fuel rod center and surface respectively and $\alpha = 0.7$.

5.6. Macroelement Geometry

The geometrical data of the macroelement are given in Table 5.23.

The water flow cross section of the macroelement does not include the water gaps between the four real-BWR elements of the macroelement.

5.7. Thermophysical Properties

Same as in section 2.7.

5.8. Neutronics-Thermohydraulics Coupling

The thermohydraulics affects the neutronics through the water density and the Doppler temperature in each neutronic mesh.

The neutronics affects the thermohydraulics through the heat sources in the fuel and in the coolant. A radially flat profile is assumed for the volumetric power density inside the fuel rod.

5.9. Operation Data

At the beginning of transient, the reactor is in equilibrium. The steady state operation data are given in Table 5.24. The inlet mass flow through the core must be properly distributed to obtain the same pressure drop across the whole core.

5.10. Heat Exchange Correlations

The conductance of the helium-filled gap between fuel and cladding (k_{gap}) is assumed

to be constant:

$$k_{gap} = 3.5 \cdot 10^3 \text{ W/m}^2 \text{ } ^\circ\text{K}$$

5.11. Pressure Drops

The inlet orifice diameter is reduced in the peripheral core macroelements shown in Figure 5.11. The pertinent inlet pressure drops vs. flow rate for standard and peripheral macroelements are shown in the Figures 5.12 and 5.13 respectively.

The frictional pressure drop inside a channel is given by the formula

$$\frac{\partial p}{\partial z} = 2 \cdot 10^{-6} G^2 f(x)$$

where:

- $\partial p / \partial z$ is the frictional pressure gradient (bar/cm);
- G is the element mass flow rate (Kg/s);
- x is the steam quality;
- $f(x)$ is the frictional factor given in Table 5.25.

The total pressure drop in a channel can be obtained by adding the inlet pressure drop to the frictional one.

6. BWR Problems

6.0. Nature of the problems

Two classes of problems (cold water injection into the core; core pressurization) are submitted for 3-D and 1-D solutions. In both cases, the initiating event is generated out of the core and involves the whole core.

The problems are to some extent complementary, as each type tends to emphasize neutronics or thermohydraulics aspects of core dynamics, mainly due to the different time scales. The core pressurization, which may be due to blockage of main steam isolation valve, induces instantaneous void collapsing with the pertinent reactivity effect, while the thermal feedback is slower. On the contrary, the cold water injection, which may be due to increase of cold feedwater flow rate or to failure of preheaters, induces void collapsing during a relatively long time (some seconds), due to the thermal inertia and to the effective water flow rate. As a consequence, neutronic and thermal responses are practically simultaneous.

6.1. Calculation of the Initial Steady State

The calculated effective multiplication factor, k_{eff} , will be used to divide the number ν of neutron produced per fission, in order to obtain a critical steady state. As a consequence, the macroscopic cross sections and their derivatives given in Tables 5.4 through 5.22 will be used only during the steady state calculations.

6.2. Transient Calculation

The cases proposed in the frame of the benchmark are:

Case D1:

Inlet cold water transient. The inlet water enthalpy vs.time is given in Fig.6.1.

Case E1: Core pressurization. The system pressure vs.time is given in Fig.6.2.

Due to the procedure adopted to establish the initial criticality, the values of $v\Sigma_{f1}$ and $v\Sigma_{f2}$ (and of their derivatives) to be used in the transient calculations will be those of Tables 5.4 through 5.22, divided by the k_{eff} calculated for the pertinent steady-state core.

The inlet mass flow through the core is constant during the transient.

7. BWR Problems: Output Requested

Results should be presented on paper and diskette in the following form (some more details are given in Section 8):

A) specification of case (case D1 or case E1)

B) "steady state results"

B1) "k-eff:"

B2) "radial power distribution at middle core:" (normalized to the peak value equal to 1: see Figure 8.5)

B3) "coolant outlet density distribution:" (see Figure 8.5)

B4) "maximum power peaking factor:"

B5) "position of maximum power peaking factor:"

B6) "axial power distribution:" (core averaged; normalized to the peak value equal to 1.: see Figure 8.4)

C) "transient results"

C1) "core power versus time:" (normalized to the steady state power; 0-20 s)

C2) "core averaged fuel temperature versus time:" (0-20 s)

C3) "maximum fuel temperature versus time:" (0-20 s)

C4) "coolant outlet density versus time:" (core averaged; 0-20 s)

C5) "radial distribution of power at layer 6:" (of the active core: see Figures 8.4- 8.5)

C5-1) " at time of power maximum:" (normalized to the peak value equal to 1)

C5-2) "at final time t = 20 s:" (normalized to the peak value equal to 1)

C7) "coolant outlet density distribution:" (see Figure 8.5)

C7-1) " at time of power maximum:"

C7-2) " at final time t = 20 s:"

8. Common Remarks for Output Format

1) Keywords delimited by apostrophes in Section 4 and in Section 7 should be written on paper and diskette as indicated.

2) Figures 8.1 and 8.4 should be used as forms for presentation on paper of axial power distributions for PWR and BWR problems respectively. The corresponding data on diskette should be ordered from the bottom to the top of the active core. Axial reflectors should not be included in the contributed results.

3) Figures 8.2, 8.3 and 8.5 should be used as forms for presentation on paper of the radial distributions of the requested quantities (power distribution at an axial layer, coolant outlet density distribution, etc.). The corresponding data on diskette should be ordered by rows from left to right and from top to bottom of the indicated forms. Radial reflector should not be included in the contributed results.

4) The time histories on paper and diskette should be presented as n ordered pairs t_j, v_j ($j=1,2,\dots,n$), where $t_j < t_{j+1}$ is the j -th time instant (in s) and v_j the corresponding value of the quantity under consideration: the number n is chosen by each contributor according to the really used model.

5) A plot of the time histories would be very useful for a first quick comparison of the transient results.

List of tables

A) Pressurized Water Reactor

- Table 2.1. - Velocity and energy release of prompt neutrons
- Table 2.2. - Decay constant and fractions of delayed neutrons
- Table 2.3. - Definition of compositions
- Table 2.4. - Key to macroscopic cross sections tables
- Table 2.5. - Cross sections $\Delta\Sigma_{CA}$ of control assemblies
- Table 2.6.1. - Cross sections and their derivatives for composition number 1
- Table 2.6.2. - Cross sections and their derivatives for composition number 2
- Table 2.6.3. - Cross sections and their derivatives for composition number 3
- Table 2.6.4. - Cross sections and their derivatives for composition number 4
- Table 2.6.5. - Cross sections and their derivatives for composition number 5
- Table 2.6.6. - Cross sections and their derivatives for composition number 6
- Table 2.6.7. - Cross sections and their derivatives for composition number 7
- Table 2.6.8. - Cross sections and their derivatives for composition number 8
- Table 2.6.9. - Cross sections and their derivatives for composition number 9
- Table 2.6.10. - Cross sections and their derivatives for composition number 10
- Table 2.6.11. - Cross sections and their derivatives for composition number 11
- Table 2.7. - Data of the subassembly (FA) geometry
- Table 2.8. - Steady state operation data

B) Boiling Water Reactor

- Table 5.1. - Prompt neutron general data
- Table 5.2. - Delayed neutron parameters
- Table 5.3. - Key to macroscopic cross section tables
- Table 5.4. - BWR: generalized nuclear composition number 1
- Table 5.5. - BWR: generalized nuclear composition number 2
- Table 5.6. - BWR: generalized nuclear composition number 3

Table 5.7. - BWR: generalized nuclear composition number 4
Table 5.8. - BWR: generalized nuclear composition number 5
Table 5.9. - BWR: generalized nuclear composition number 6
Table 5.10. - BWR: generalized nuclear composition number 7
Table 5.11. - BWR: generalized nuclear composition number 8
Table 5.12. - BWR: generalized nuclear composition number 9
Table 5.13. - BWR: generalized nuclear composition number 10
Table 5.14. - BWR: generalized nuclear composition number 11
Table 5.15. - BWR: generalized nuclear composition number 12
Table 5.16. - BWR: generalized nuclear composition number 13
Table 5.17. - BWR: generalized nuclear composition number 14
Table 5.18. - BWR: generalized nuclear composition number 15
Table 5.19. - BWR: generalized nuclear composition number 16
Table 5.20. - BWR: generalized nuclear composition number 17
Table 5.21. - BWR: generalized nuclear composition number 18
Table 5.22. - BWR: generalized nuclear composition number 19
Table 5.23. - Data of macroelement geometry
Table 5.24. - Steady state operation data
Table 5.25. - Friction factor vs. steam quality

List of figures

A) Pressurized Water Reactor

- Fig.2.1. - Cross section of the reactor core
- Fig.2.2. - Arrangement of control assemblies
- Fig.2.3. - Composition numbers in axial layers 1 and 18 (bottom and top reflector)
- Fig.2.4. - Composition numbers in axial layer 2 (bottom layer of active core)
- Fig.2.5. - Composition numbers in axial layers 3 through 17 (active core)
- Fig.3.1. - Case A1: Initial configuration of control assemblies
- Fig.3.2. - Case A2: Initial configuration of control assemblies
- Fig.3.3. - Case B1: Initial configuration of control assemblies
- Fig.3.4. - Case B2: Initial configuration of control assemblies
- Fig.3.5. - Case C1: Initial configuration of control assemblies
- Fig.3.6. - Case C2: Initial configuration of control assemblies
- Fig.8.1. - Form for power axial distribution
- Fig.8.2. - Form for power radial distribution in Cases A1-A2-B1-B2
- Fig.8.3. - Form for power radial distribution in Cases C1-C2

B) Boiling Water Reactor

- Fig.5.1. - BWR initial map
- Fig.5.2. - BWR macroelement type 1
- Fig.5.3. - BWR macroelement type 2
- Fig.5.4. - BWR macroelement type 3
- Fig.5.5. - BWR macroelement type 4
- Fig.5.6. - BWR macroelement type 5
- Fig.5.7. - BWR macroelement type 6
- Fig.5.8. - BWR macroelement type 7
- Fig.5.9. - BWR macroelement type 8
- Fig.5.10. - BWR macroelement type 9
- Fig.5.11. - Map of macroelement inlet orifices
- Fig.5.12. - Inlet pressure drop vs. flow rate (standard macroelement)

Fig.5.13. - Inlet pressure drop vs. flow rate (peripheral macroelement)

Fig.6.1. - Case D: inlet water subcooling vs. time

Fig.6.2. - Case E: core pressure vs. time

Fig.8.4. - Form for axial distributions

Fig.8.5. - Form for radial distributions in Cases D1-E1

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3-D LWR CORE TRANSIENT BENCHMARK
Pressurized Water Reactor

	fast energy group	thermal energy group
neutron velocity (cm/s)	$0.28 \cdot 10^6$	$0.44 \cdot 10^6$
energy release (Ws/fission)	$0.3213 \cdot 10^{-10}$	$0.3206 \cdot 10^{-10}$

Table 2.1 - Velocities and energy release of prompt neutrons

group	decay constant (s ⁻¹)	relative fraction of delayed neutrons
1	0.0128	0.034
2	0.0318	0.200
3	0.1190	0.183
4	0.3181	0.404
5	1.4027	0.145
6	3.9286	0.034
total fraction of delayed neutrons:		0.76 %

Table 2.2 - Decay constant and fractions of delayed neutrons

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Pressurized Water Reactor

composition number	characteristics
1	axial reflector
2	radial reflector
3	radial reflector re-entrant corner
4	2.1 w/o
5	2.6 w/o
6	3.1 w/o
7	2.6 w/o, 12 burnable absorbers rods (BA)
8	2.6 w/o, 16 BA
9	2.6 w/o, 20 BA
10	3.1 w/o, 12 BA
11	3.1 w/o, 16 BA

Table 2.3 - Definition of compositions

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Pressurized Water Reactor

$\Sigma_{tr,1}$	$\Sigma_{1 \rightarrow 2}$	$\Sigma_{a,1}$	$\nu \Sigma_{f,1}$	$\Sigma_{f,1}$
$\Sigma_{tr,2}$	$\Sigma_{a,2}$	$\nu \Sigma_{f,2}$	$\Sigma_{f,2}$	Comp.Nr.
$\partial \Sigma_{tr,1} / \partial c$	$\partial \Sigma_{1 \rightarrow 2} / \partial c$	$\partial \Sigma_{a,1} / \partial c$	$\partial \nu \Sigma_{f,1} / \partial c$	$\partial \Sigma_{f,1} / \partial c$
$\partial \Sigma_{tr,2} / \partial c$	$\partial \Sigma_{a,2} / \partial c$	$\partial \nu \Sigma_{f,2} / \partial c$	$\partial \Sigma_{f,2} / \partial c$	c_0
$\partial \Sigma_{tr,1} / \partial T_M$	$\partial \Sigma_{1 \rightarrow 2} / \partial T_M$	$\partial \Sigma_{a,1} / \partial T_M$	$\partial \nu \Sigma_{f,1} / \partial T_M$	$\partial \Sigma_{f,1} / \partial T_M$
$\partial \Sigma_{tr,2} / \partial T_M$	$\partial \Sigma_{a,2} / \partial T_M$	$\partial \nu \Sigma_{f,2} / \partial T_M$	$\partial \Sigma_{f,2} / \partial T_M$	T_{M0}
$\partial \Sigma_{tr,1} / \partial \rho$	$\partial \Sigma_{1 \rightarrow 2} / \partial \rho$	$\partial \Sigma_{a,1} / \partial \rho$	$\partial \nu \Sigma_{f,1} / \partial \rho$	$\partial \Sigma_{f,1} / \partial \rho$
$\partial \Sigma_{tr,2} / \partial \rho$	$\partial \Sigma_{a,2} / \partial \rho$	$\partial \nu \Sigma_{f,2} / \partial \rho$	$\partial \Sigma_{f,2} / \partial \rho$	ρ_0
$\partial \Sigma_{tr,1} / \partial \sqrt{T_F}$	$\partial \Sigma_{1 \rightarrow 2} / \partial \sqrt{T_F}$	$\partial \Sigma_{a,1} / \partial \sqrt{T_F}$	$\partial \nu \Sigma_{f,1} / \partial \sqrt{T_F}$	$\partial \Sigma_{f,1} / \partial \sqrt{T_F}$
$\partial \Sigma_{tr,2} / \partial \sqrt{T_F}$	$\partial \Sigma_{a,2} / \partial \sqrt{T_F}$	$\partial \nu \Sigma_{f,2} / \partial \sqrt{T_F}$	$\partial \Sigma_{f,2} / \partial \sqrt{T_F}$	T_{F0}

where:

- Comp.Nr. is the composition number, ranging from 1 to 11
- c is the boron concentration (ppm)
- ρ is the water density (g/cm^3);
- T_M is the moderator temperature ($^{\circ}C$);
- T_F is the Doppler temperature ($^{\circ}C$); *(use kelvin in formula)*

Reference values are labeled with subscript 0. Macroscopic cross sections are in units of cm^{-1} . The meanings of the indices of cross sections are:

1,2	fast or thermal neutron group
tr	transport
1 $\rightarrow$ 2	scattering from group 1 into group 2
a	absorption
f	fission
ν	number of neutrons per fission

The transport cross section is related to the diffusion constant D by $D = 1/(3 \Sigma_{tr})$

Table 2.4 - Key to macroscopic cross section tables

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0.373220E-02	-0.319253E-02	0.247770E-02	-0.102786E-03	-0.377989E-04
-0.219926E-01	0.255875E-01	-0.282319E-02	-0.115483E-02	absorber type 1
0.374092E-02	-0.314239E-02	0.242926E-02	-0.122634E-03	-0.459250E-04
-0.167503E-01	0.256478E-01	-0.328086E-02	-0.134262E-02	absorber type 2
0.697102E-02	-0.119034E-02	0.879034E-04	-0.655496E-04	-0.197926E-04
-0.113498E-01	0.170043E-02	-0.146252E-02	-0.599154E-03	driver

Table 2.5 - Cross sections $\Delta\Sigma_{CA}$ of control assemblies

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0.532058E-01	0.264554E-01	0.373279E-03	0.000000E-00	0.000000E-00
0.386406E-00	0.177215E-01	0.000000E-00	0.000000E-00	1
0.611833E-07	0.791457E-09	0.187731E-06	0.000000E-00	0.000000E-00
0.517535E-05	0.102635E-04	0.000000E-00	0.000000E-00	1200.2
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	306.6
0.745756E-01	0.371310E-01	0.207688E-03	0.000000E-00	0.000000E-00
0.533634E-00	0.758421E-02	0.000000E-00	0.000000E-00	0.7125
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	618.3

Table 2.6.1 - Cross sections and their derivatives for composition number 1

0.295609E-00	0.231613E-01	0.118782E-02	0.000000E-00	0.000000E-00
0.245931E+01	0.252618E-00	0.000000E-00	0.000000E-00	2
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.776184E-03	0.844695E-04	0.000000E-00	0.000000E-00	1200.2
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	306.6
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.7125
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	618.3

Table 2.6.2 - Cross sections and their derivatives for composition number 2

0.295609E-00	0.200808E-01	0.118782E-02	0.000000E-00	0.000000E-00
0.245931E+01	0.252618E-00	0.000000E-00	0.000000E-00	3
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.776184E-03	0.844695E-04	0.000000E-00	0.000000E-00	1200.2
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	306.6
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.7125
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00
0.000000E-00	0.000000E-00	0.000000E-00	0.000000E-00	618.3

Table 2.6.3 - Cross sections and their derivatives for composition number 3

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0.222117E-00	0.182498E-01	0.871774E-02	0.498277E-02	0.190224E-02
0.803140E-00	0.652550E-01	0.839026E-01	0.343581E-01	4
0.347809E-07	-0.108590E-06	0.128505E-06	-0.112099E-08	-0.548361E-09
-0.976510E-05	0.708807E-05	-0.243045E-05	-0.995273E-06	1200.2
-0.203310E-05	0.809676E-06	0.212191E-06	0.124709E-06	0.445177E-07
-0.108674E-03	-0.315597E-04	-0.416439E-04	-0.170531E-04	306.6
0.135665E-00	0.293195E-01	0.155185E-02	0.920694E-03	0.318680E-03
0.992628E-00	0.252662E-01	0.247746E-01	0.101452E-01	0.7125
-0.309197E-04	-0.275536E-04	0.349709E-04	0.640134E-06	0.222662E-06
-0.137292E-03	-0.371806E-04	-0.563037E-04	-0.230564E-04	618.3

Table 2.6.4 - Cross sections and their derivatives for composition number 4

0.221914E-00	0.180040E-01	0.906133E-02	0.557659E-02	0.214498E-02
0.795538E-00	0.723354E-01	0.998629E-01	0.408938E-01	5
0.353826E-07	-0.106951E-06	0.126709E-06	-0.167880E-08	-0.777980E-09
-0.850169E-05	0.682311E-05	-0.272445E-05	-0.111566E-05	1200.2
-0.198080E-05	0.858474E-06	0.226000E-06	0.135145E-06	0.488317E-07
-0.906150E-04	-0.321435E-04	-0.453102E-04	-0.185545E-04	306.6
0.135748E-00	0.292696E-01	0.161491E-02	0.964160E-03	0.336573E-03
0.981985E-00	0.286667E-01	0.314993E-01	0.128990E-01	0.7125
-0.308607E-04	-0.276766E-04	0.351798E-04	0.997431E-06	0.369389E-06
-0.117481E-03	-0.377039E-04	-0.604155E-04	-0.247402E-04	618.3

Table 2.6.5 - Cross sections and their derivatives for composition number 5

0.221715E-00	0.177670E-01	0.938496E-02	0.615047E-02	0.237972E-02
0.789253E-00	0.789203E-01	0.114667E-00	0.469561E-01	6
0.359838E-07	-0.105374E-06	0.124986E-06	-0.221038E-08	-0.996653E-09
-0.746251E-05	0.659798E-05	-0.295883E-05	-0.121164E-05	1200.2
-0.192434E-05	0.903494E-06	0.239939E-06	0.149084E-06	0.545978E-07
-0.762786E-04	-0.323776E-04	-0.478475E-04	-0.195937E-04	306.6
0.135827E-00	0.292154E-01	0.168015E-02	0.101410E-02	0.357207E-03
0.972267E-00	0.319571E-01	0.381097E-01	0.156060E-01	0.7125
-0.309165E-04	-0.278390E-04	0.353841E-04	0.141847E-05	0.542384E-06
-0.101337E-03	-0.377558E-04	-0.630960E-04	-0.258379E-04	618.3

Table 2.6.6 - Cross sections and their derivatives for composition number 6

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0.222039E-00	0.171381E-01	0.931692E-02	0.555010E-02	0.213629E-02
0.776230E-00	0.796328E-01	0.985576E-01	0.403596E-01	7
0.337806E-07	-0.100873E-06	0.119869E-06	-0.171323E-08	-0.777980E-09
-0.673744E-05	0.629310E-05	-0.255359E-05	-0.104561E-05	1200.2
-0.269634E-05	0.701311E-06	0.248530E-06	0.140773E-06	0.488317E-07
-0.762435E-04	-0.300119E-04	-0.420202E-04	-0.172074E-04	306.6
0.131033E-00	0.282489E-01	0.168397E-02	0.981951E-03	0.336573E-03
0.934697E-00	0.314240E-01	0.351588E-01	0.144016E-01	0.7125
-0.313746E-04	-0.273550E-04	0.348699E-04	0.945431E-06	0.369389E-06
-0.108271E-03	-0.372748E-04	-0.579662E-04	-0.237320E-04	618.3

Table 2.6.7 - Cross sections and their derivatives for composition number 7

0.222083E-00	0.168501E-01	0.940032E-02	0.554083E-02	0.213318E-02
0.769969E-00	0.821087E-01	0.980059E-01	0.401338E-01	8
0.332495E-07	-0.988578E-07	0.117585E-06	-0.172421E-08	-0.793325E-09
-0.619725E-05	0.611904E-05	-0.248880E-05	-0.101904E-05	1200.2
-0.307905E-05	0.617380E-06	0.261854E-06	0.143235E-06	0.522555E-07
-0.733397E-04	-0.291929E-04	-0.407701E-04	-0.166956E-04	306.6
0.129379E-00	0.278895E-01	0.171972E-02	0.988437E-03	0.346474E-03
0.918171E-00	0.324715E-01	0.363251E-01	0.148808E-01	0.7125
-0.315503E-04	-0.272381E-04	0.347274E-04	0.926078E-06	0.339044E-06
-0.105521E-03	-0.371808E-04	-0.571108E-04	-0.233804E-04	618.3

Table 2.6.8 - Cross sections and their derivatives for composition number 8

0.222127E-00	0.165626E-01	0.948286E-02	0.553137E-02	0.213003E-02
0.763813E-00	0.845912E-01	0.974109E-01	0.398902E-01	9
0.327201E-07	-0.968489E-07	0.115319E-06	-0.173502E-08	-0.796917E-09
-0.568220E-05	0.594711E-05	-0.242240E-05	-0.991817E-06	1200.2
-0.353877E-05	0.516547E-06	0.274313E-06	0.146019E-06	0.534283E-07
-0.713711E-04	-0.283041E-04	-0.394319E-04	-0.161475E-04	306.6
0.127682E-00	0.275202E-01	0.174989E-02	0.995175E-03	0.349236E-03
0.901293E-00	0.335945E-01	0.374499E-01	0.153431E-01	0.7125
-0.317281E-04	-0.271169E-04	0.346026E-04	0.905802E-06	0.330425E-06
-0.102525E-03	-0.370201E-04	-0.561543E-04	-0.229872E-04	618.3

Table 2.6.9 - Cross sections and their derivatives for composition number 9

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0.221836E-00	0.169043E-01	0.963720E-02	0.612382E-02	0.237097E-02
0.770705E-00	0.861187E-01	0.113241E-00	0.463724E-01	10
0.343859E-07	- 0.993312E-07	0.118186E-06	- 0.224335E-08	- 0.996653E-09
- 0.586898E-05	0.608443E-05	- 0.277657E-05	- 0.113696E-05	1200.2
- 0.263907E-05	0.744320E-06	0.264289E-06	0.155858E-06	0.545978E-07
- 0.639554E-04	- 0.303509E-04	- 0.444431E-04	- 0.181997E-04	306.6
0.131116E-00	0.281877E-01	0.175528E-02	0.103522E-02	0.357207E-03
0.924925E-00	0.349853E-01	0.420693E-01	0.172298E-01	0.7125
- 0.314192E-04	- 0.275049E-04	0.350637E-04	0.135642E-05	0.542384E-06
- 0.938886E-04	- 0.371403E-04	- 0.605052E-04	- 0.247739E-04	618.3

Table 2.6.10 - Cross sections and their derivatives for composition number 10

0.221878E-00	0.166175E-01	0.971937E-02	0.611444E-02	0.236781E-02
0.764704E-00	0.885488E-01	0.112635E-00	0.461246E-01	11
0.338559E-07	- 0.973291E-07	0.115917E-06	- 0.225369E-08	- 0.101112E-08
- 0.538345E-05	0.591697E-05	- 0.270780E-05	- 0.110878E-05	1200.2
- 0.302147E-05	0.659521E-06	0.279060E-06	0.158814E-06	0.586765E-07
- 0.616984E-04	- 0.295626E-04	- 0.431588E-04	- 0.176738E-04	306.6
0.129463E-00	0.278259E-01	0.179499E-02	0.104291E-02	0.368921E-03
0.908456E-00	0.361032E-01	0.433215E-01	0.177435E-01	0.7125
- 0.315908E-04	- 0.273835E-04	0.349119E-04	0.133336E-05	0.506596E-06
- 0.917126E-04	- 0.369909E-04	- 0.596284E-04	- 0.244141E-04	618.3

Table 2.6.11 - Cross sections and their derivatives for composition number 11

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Pellet diameter	8.239 mm
Clad diameter (outside)	9.517 mm
Clad wall thickness	0.571 mm
FR pitch	12.655 mm
Guide tube diameter (outside)	12.259 mm
Guide tube diameter (inside)	11.448 mm
Geometry	17 x 17 - 25
Number of fuel pins	264
Number of guide tubes	25

Table 2.7 - Data of the subassembly (FA) geometry

Core thermal output	2775	MW
Core inlet temperature	286	°C
Core pressure	155	bar
Net mass flow through core	12893	Kg/s

Table 2.8 - Steady state operation data

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Fast neutron inverse velocity	v_1^{-1}	$3.57 \cdot 10^{-8} \text{ cm}^{-1} \text{ s}$
Thermal neutron inverse velocity	v_2^{-1}	$2.27 \cdot 10^{-6} \text{ cm}^{-1} \text{ s}$
Prompt energy release per fission	E_0	$3.20 \cdot 10^{-11} \text{ J/fission}$

Table 5.1 - Prompt neutron general data

i	$\lambda_i \text{ (s}^{-1}\text{)}$	β_i
1	0.013	0.00026
2	0.032	0.00152
3	0.119	0.00139
4	0.318	0.00307
5	1.403	0.00110
6	3.929	0.00026
Total β		0.00760

Table 5.2 - Delayed neutron parameters

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Boiling Water Reactor

$\Sigma_{t,1}$	$\Sigma_{1 \rightarrow 2}$	$\Sigma_{a,1}$	$v\Sigma_{f,1}$	$\Sigma_{f,1}$
$\Sigma_{t,2}$	$\Sigma_{a,2}$	$v\Sigma_{f,2}$	$\Sigma_{f,2}$	C.Id.
$\partial\Sigma_{t,1}/\partial\rho$	$\partial\Sigma_{1 \rightarrow 2}/\partial\rho$	$\partial\Sigma_{a,1}/\partial\rho$	$\partial v\Sigma_{f,1}/\partial\rho$	$\partial\Sigma_{f,1}/\partial\rho$
$\partial\Sigma_{t,2}/\partial\rho$	$\partial\Sigma_{a,2}/\partial\rho$	$\partial v\Sigma_{f,2}/\partial\rho$	$\partial\Sigma_{f,2}/\partial\rho$	ρ_0
$\partial\Sigma_{t,1}/\partial\sqrt{T}$	$\partial\Sigma_{1 \rightarrow 2}/\partial\sqrt{T}$	$\partial\Sigma_{a,1}/\partial\sqrt{T}$	$\partial v\Sigma_{f,1}/\partial\sqrt{T}$	$\partial\Sigma_{f,1}/\partial\sqrt{T}$
$\partial\Sigma_{t,2}/\partial\sqrt{T}$	$\partial\Sigma_{a,2}/\partial\sqrt{T}$	$\partial v\Sigma_{f,2}/\partial\sqrt{T}$	$\partial\Sigma_{f,2}/\partial\sqrt{T}$	T_0

where:

- ρ is the water density (g/cm³);
- T is the Doppler temperature (°K);
- ρ_0 is the reference water density (g/cm³);
- T_0 is the reference Doppler temperature (°K);
- C.Id. is the Composition Identifier, ranging from 1 to 19
- the cross sections are expressed in cm⁻¹

Table 5.3 - Key to macroscopic cross section tables

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3-D LWR CORE TRANSIENT BENCHMARK

BOILING WATER REACTOR

0.111030E+00	0.225950E-01	0.392000E-03	0.000000E+00	0.000000E+00	
0.830012E+00	0.148010E-01	0.000000E+00	0.000000E+00		1
0.219320E-06	0.219320E-06	0.856719E-09	0.000000E+00	0.000000E+00	
0.701825E-05	0.109660E-06	0.000000E+00	0.000000E+00	0.850000E+00	
0.000000E+00	0.000000E-00	0.000000E-00	0.000000E+00	0.000000E+00	
0.000000E+00	0.000000E-00	0.000000E-00	0.000000E+00	5.731500E+02	

Tab.5.4 - BWR : generalized nuclear composition number 1

0.189784E+00	0.141764E-01	0.102352E-01	0.446986E-02	0.181431E-02	
0.694136E+00	0.749127E-01	0.828220E-01	0.336177E-01		2
0.130164E+00	0.196627E-01	0.238730E-02	0.111084E-02	0.464362E-03	
0.759141E+00	0.126336E-01	0.246360E-01	0.102472E-01	0.550000E+00	
0.000000E+00	-.160580E-04	0.200902E-04	0.000000E+00	0.000000E+00	
-.823459E-04	-.262873E-04	-.350770E-04	-.142416E-04	5.731500E+02	

Tab.5.5 - BWR : generalized nuclear composition number 2

0.188554E+00	0.142295E-01	0.103417E-01	0.446539E-02	0.181220E-02	
0.693963E+00	0.768592E-01	0.804386E-01	0.326449E-01		3
0.128750E+00	0.197654E-01	0.247360E-02	0.111093E-02	0.464733E-03	
0.752907E+00	0.127195E-01	0.236482E-01	0.984506E-02	0.550000E+00	
0.000000E+00	-.161574E-04	0.201801E-04	0.000000E+00	0.000000E+00	
-.825240E-04	-.270360E-04	-.341409E-04	-.138593E-04	5.731500E+02	

Tab.5.6 - BWR : generalized nuclear composition number 3

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3-D LWR CORE TRANSIENT BENCHMARK

BOILING WATER REACTOR

0.134270E+00	0.181770E-01	0.553000E-03	0.000000E+00	0.000000E+00	
0.730050E+00	0.622329E-02	0.000000E+00	0.000000E+00		4
0.175456E-05	0.219320E-06	0.000000E+00	0.000000E+00	0.000000E+00	
0.140365E-04	-.685375E-08	0.000000E+00	0.000000E+00	0.550000E+00	
0.000000E+00	0.000000E-00	0.000000E-00	0.000000E+00	0.000000E+00	
0.000000E+00	0.000000E-00	0.000000E-00	0.000000E+00	5.731500E+02	

Tab.5.7 - BWR : generalized nuclear composition number 4

0.189186E+00	0.143548E-01	0.101071E-01	0.413061E-02	0.167631E-02	
0.693475E+00	0.683185E-01	0.738611E-01	0.299759E-01		5
0.129802E+00	0.198449E-01	0.227347E-02	0.960766E-03	0.402857E-03	
0.771240E+00	0.108673E-01	0.211180E-01	0.879825E-02	0.550000E+00	
0.000000E+00	-.162970E-04	0.204046E-04	0.000000E+00	0.000000E+00	
-.824488E-04	-.239924E-04	-.313155E-04	-.127127E-04	5.731500E+02	

Tab.5.8 - BWR : generalized nuclear composition number 5

0.188264E+00	0.143946E-01	0.101869E-01	0.412726E-02	0.167472E-02	
0.693345E+00	0.697783E-01	0.720736E-01	0.292463E-01		6
0.128744E+00	0.199218E-01	0.233817E-02	0.960828E-03	0.403124E-03	
0.766559E+00	0.109320E-01	0.203775E-01	0.849679E-02	0.550000E+00	
0.000000E+00	-.163716E-04	0.204720E-04	0.000000E+00	0.000000E+00	
-.825824E-04	-.245539E-04	-.306135E-04	-.124259E-04	5.731500E+02	

Tab.5.9 - BWR : generalized nuclear composition number 6

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3-D LWR CORE TRANSIENT BENCHMARK

BOILING WATER REACTOR

0.199654E+00 0.164565E-01 0.709736E-02 0.416239E-02 0.169163E-02
 0.718647E+00 0.484724E-01 0.649081E-01 0.263796E-01 7
 0.127443E+00 0.197129E-01 0.170083E-02 0.696670E-03 0.290777E-03
 0.750235E+00 0.126452E-01 0.130177E-01 0.540898E-02 0.550000E+00
 0.000000E+00 -.189002E-04 0.231603E-04 0.000000E+00 0.000000E+00
 -.867114E-04 -.172597E-04 -.277381E-04 -.112749E-04 5.731500E+02

Tab.5.10 - BWR : generalized nuclear composition number 7

0.198692E+00 0.165185E-01 0.715434E-02 0.416026E-02 0.169070E-02
 0.719003E+00 0.490908E-01 0.636570E-01 0.258699E-01 8
 0.126783E+00 0.197951E-01 0.171473E-02 0.693127E-03 0.289441E-03
 0.746987E+00 0.125016E-01 0.124660E-01 0.518429E-02 0.550000E+00
 0.000000E+00 -.189895E-04 0.232572E-04 0.000000E+00 0.000000E+00
 -.868472E-04 -.174963E-04 -.272325E-04 -.110690E-04 5.731500E+02

Tab.5.11 - BWR : generalized nuclear composition number 8

0.189151E+00 0.143552E-01 0.101112E-01 0.413003E-02 0.167607E-02
 0.693476E+00 0.683829E-01 0.738046E-01 0.299530E-01 9
 0.129752E+00 0.198434E-01 0.228093E-02 0.959197E-03 0.402219E-03
 0.771049E+00 0.108792E-01 0.211062E-01 0.879375E-02 0.550000E+00
 0.000000E+00 -.162923E-04 0.203980E-04 0.000000E+00 0.000000E+00
 -.824204E-04 -.240056E-04 -.312795E-04 -.126980E-04 5.731500E+02

Tab.5.12 - BWR : generalized nuclear composition number 9

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3-D LWR CORE TRANSIENT BENCHMARK

BOILING WATER REACTOR

0.188381E+00	0.143893E-01	0.101774E-01	0.412816E-02	0.167512E-02	
0.693427E+00	0.696467E-01	0.723935E-01	0.293768E-01		10
0.128909E+00	0.199168E-01	0.232403E-02	0.962980E-03	0.403946E-03	
0.766905E+00	0.109351E-01	0.205383E-01	0.856259E-02	0.550000E+00	
0.000000E+00	-.163709E-04	0.204749E-04	0.000000E+00	0.000000E+00	
-.826201E-04	-.245168E-04	-.307625E-04	-.124867E-04	5.731500E+02	

Tab.5.13 - BWR : generalized nuclear composition number 10

0.187960E+00	0.144036E-01	0.102185E-01	0.412665E-02	0.167439E-02	
0.693550E+00	0.704713E-01	0.716701E-01	0.290810E-01		11
0.128450E+00	0.199429E-01	0.236219E-02	0.963165E-03	0.404145E-03	
0.762266E+00	0.110489E-01	0.204017E-01	0.850699E-02	0.550000E+00	
0.000000E+00	-.163872E-04	0.204833E-04	0.000000E+00	0.000000E+00	
-.826179E-04	-.248087E-04	-.304546E-04	-.123607E-04	5.731500E+02	

Tab.5.14 - BWR : generalized nuclear composition number 11

0.188575E+00	0.143771E-01	0.101653E-01	0.412888E-02	0.167544E-02	
0.693636E+00	0.694981E-01	0.728618E-01	0.295674E-01		12
0.129158E+00	0.198915E-01	0.231898E-02	0.963117E-03	0.403967E-03	
0.765378E+00	0.110057E-01	0.208959E-01	0.870788E-02	0.550000E+00	
0.000000E+00	-.163375E-04	0.204384E-04	0.000000E+00	0.000000E+00	
-.825289E-04	-.244344E-04	-.309227E-04	-.125519E-04	5.731500E+02	

Tab.5.15 - BWR : generalized nuclear composition number 12

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3-D LWR CORE TRANSIENT BENCHMARK

BOILING WATER REACTOR

0.189081E+00	0.143562E-01	0.101196E-01	0.412887E-02	0.167559E-02	
0.693478E+00	0.685118E-01	0.736916E-01	0.299070E-01		13
0.129650E+00	0.198407E-01	0.229584E-02	0.956030E-03	0.400945E-03	
0.770663E+00	0.109047E-01	0.210825E-01	0.878421E-02	0.550000E+00	
0.000000E+00	-.162829E-04	0.203849E-04	0.000000E+00	0.000000E+00	
-.823634E-04	-.240320E-04	-.312074E-04	-.126688E-04	5.731500E+02	

Tab.5.16 - BWR : generalized nuclear composition number 13

0.188616E+00	0.143789E-01	0.101582E-01	0.412995E-02	0.167592E-02	
0.693591E+00	0.693833E-01	0.730336E-01	0.296377E-01		14
0.129248E+00	0.199065E-01	0.229551E-02	0.967291E-03	0.405584E-03	
0.767578E+00	0.109412E-01	0.208614E-01	0.869330E-02	0.550000E+00	
0.000000E+00	-.163695E-04	0.204806E-04	0.000000E+00	0.000000E+00	
-.826954E-04	-.244425E-04	-.310605E-04	-.126082E-04	5.731500E+02	

Tab.5.17 - BWR : generalized nuclear composition number 14

0.187354E+00	0.144216E-01	0.102817E-01	0.412543E-02	0.167370E-02	
0.693960E+00	0.718573E-01	0.708631E-01	0.287502E-01		15
0.127871E+00	0.199848E-01	0.241000E-02	0.967730E-03	0.406160E-03	
0.753668E+00	0.112829E-01	0.204502E-01	0.852695E-02	0.550000E+00	
0.000000E+00	-.164185E-04	0.205059E-04	0.000000E+00	0.000000E+00	
-.826889E-04	-.253184E-04	-.301369E-04	-.122304E-04	5.731500E+02	

Tab.5.18 - BWR : generalized nuclear composition number 15

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3-D LWR CORE TRANSIENT BENCHMARK

BOILING WATER REACTOR

0.199871E+00	0.164480E-01	0.708533E-02	0.415877E-02	0.169022E-02	
0.722345E+00	0.482128E-01	0.650624E-01	0.264432E-01		16
0.127402E+00	0.197119E-01	0.169762E-02	0.692305E-03	0.289009E-03	
0.738630E+00	0.127121E-01	0.127594E-01	0.530426E-02	0.550000E+00	
0.000000E+00	-.188723E-04	0.231300E-04	0.000000E+00	0.000000E+00	
-.866561E-04	-.171504E-04	-.277728E-04	-.112893E-04	5.731500E+02	

Tab.5.19 - BWR : generalized nuclear composition number 16

0.198410E+00	0.165210E-01	0.717808E-02	0.416892E-02	0.169414E-02	
0.718597E+00	0.500243E-01	0.643471E-01	0.261493E-01		17
0.126972E+00	0.197836E-01	0.172207E-02	0.704641E-03	0.294266E-03	
0.745357E+00	0.126051E-01	0.137254E-01	0.569860E-02	0.550000E+00	
0.000000E+00	-.189906E-04	0.232614E-04	0.000000E+00	0.000000E+00	
-.867723E-04	-.178265E-04	-.277559E-04	-.112813E-04	5.731500E+02	

Tab.5.20 - BWR : generalized nuclear composition number 17

0.197215E+00	0.165982E-01	0.725399E-02	0.416619E-02	0.169287E-02	
0.719295E+00	0.509656E-01	0.626218E-01	0.254457E-01		18
0.126127E+00	0.198932E-01	0.173860E-02	0.701297E-03	0.293073E-03	
0.741955E+00	0.125542E-01	0.126783E-01	0.527268E-02	0.550000E+00	
0.000000E+00	-.191256E-04	0.234010E-04	0.000000E+00	0.000000E+00	
-.870674E-04	-.182100E-04	-.268636E-04	-.109176E-04	5.731500E+02	

Tab.5.21 - BWR : generalized nuclear composition number 18

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BOILING WATER REACTOR

0.184542E+00	0.375790E-01	0.359000E-03	0.000000E+00	0.000000E+00	
0.136864E+01	0.108680E-01	0.000000E+00	0.000000E+00		19
0.000000E+00	0.109660E-06	0.428360E-09	0.000000E+00	0.000000E+00	
0.175456E-05	0.109660E-06	0.000000E+00	0.000000E+00	0.766370E+00	
0.000000E+00	0.000000E-00	0.000000E-00	0.000000E+00	0.000000E+00	
0.000000E+00	0.000000E-00	0.000000E-00	0.000000E+00	5.731500E+02	

Tab.5.22 - BWR : generalized nuclear composition number 19

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Boiling Water Reactor

Number of fuel rods	196
Outer clad diameter	1.430 cm
Inner clad diameter	1.267 cm
Pellet diameter	1.237 cm
Fuel rod pitch	1.875 cm
Flow cross-section	400.78 cm ²
Heated perimeter	880.5256 cm
Hydraulic diameter	1.4730 cm

Tab. 5.23 - Data of macroelement geometry

Core thermal power	1800 MW
Total inlet mass flow rate	13000 Kg/s
Core pressure	67.0 bar
Coolant inlet subcooling	46.52 KJ/Kg

Tab. 5.24 - Steady state operation data

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Boiling Water Reactor

x	$f'(x)$	x	$f'(x)$	x	$f'(x)$
0.00	1.00	0.15	7.25	0.30	11.75
0.01	1.70	0.16	7.55	0.31	12.00
0.02	2.30	0.17	7.85	0.32	12.25
0.03	2.80	0.18	8.15	0.33	12.50
0.04	3.20	0.19	8.45	0.34	12.75
0.05	3.60	0.20	8.75	0.35	13.00
0.06	4.00	0.21	9.05	0.36	13.25
0.07	4.40	0.22	9.35	0.37	13.50
0.08	4.80	0.23	9.65	0.38	13.75
0.09	5.15	0.24	9.95	0.39	14.00
0.10	5.50	0.25	10.25	0.40	14.25
0.11	5.85	0.26	10.55	0.41	14.50
0.12	6.20	0.27	10.85	0.42	14.75
0.13	6.55	0.28	11.15	0.43	15.00
0.14	6.90	0.29	11.45	0.44	15.25

x	$f'(x)$	x	$f'(x)$
0.45	15.50	0.60	19.25
0.46	15.75	0.61	19.50
0.47	16.00	0.62	19.75
0.48	16.25	0.63	20.00
0.49	16.50	0.64	20.25
0.50	16.75	0.65	20.50
0.51	17.00	0.66	20.75
0.52	17.25	0.67	21.00
0.53	17.50	0.68	21.20
0.54	17.75	0.69	21.40
0.55	18.00	0.70	21.60
0.56	18.25	0.71	21.75
0.57	18.50	0.72	21.90
0.58	18.75	0.73	22.05
0.59	19.00	0.74	22.30

Table 5.25 - Friction factor vs. steam quality

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Pressurized Water Reactor

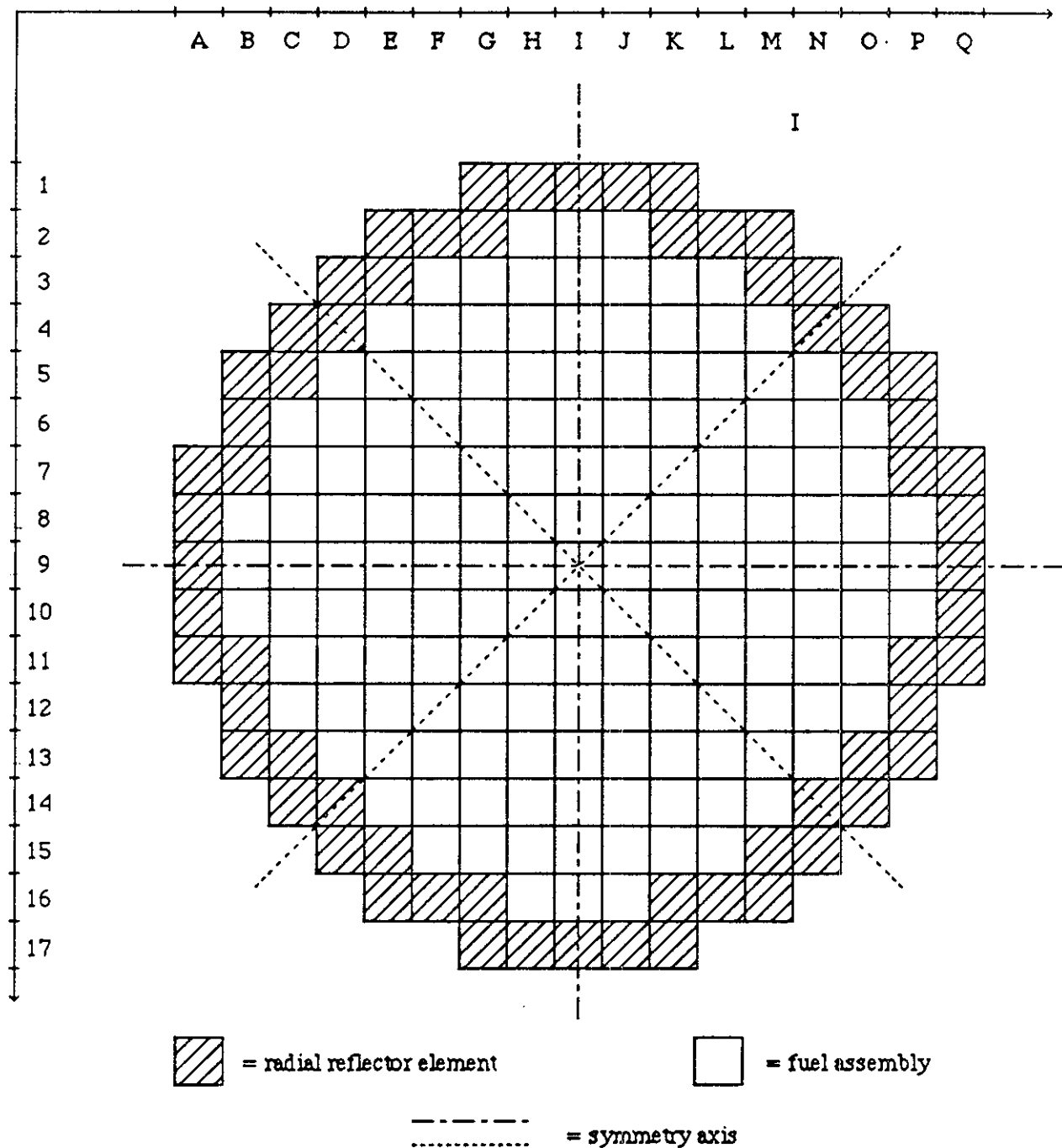
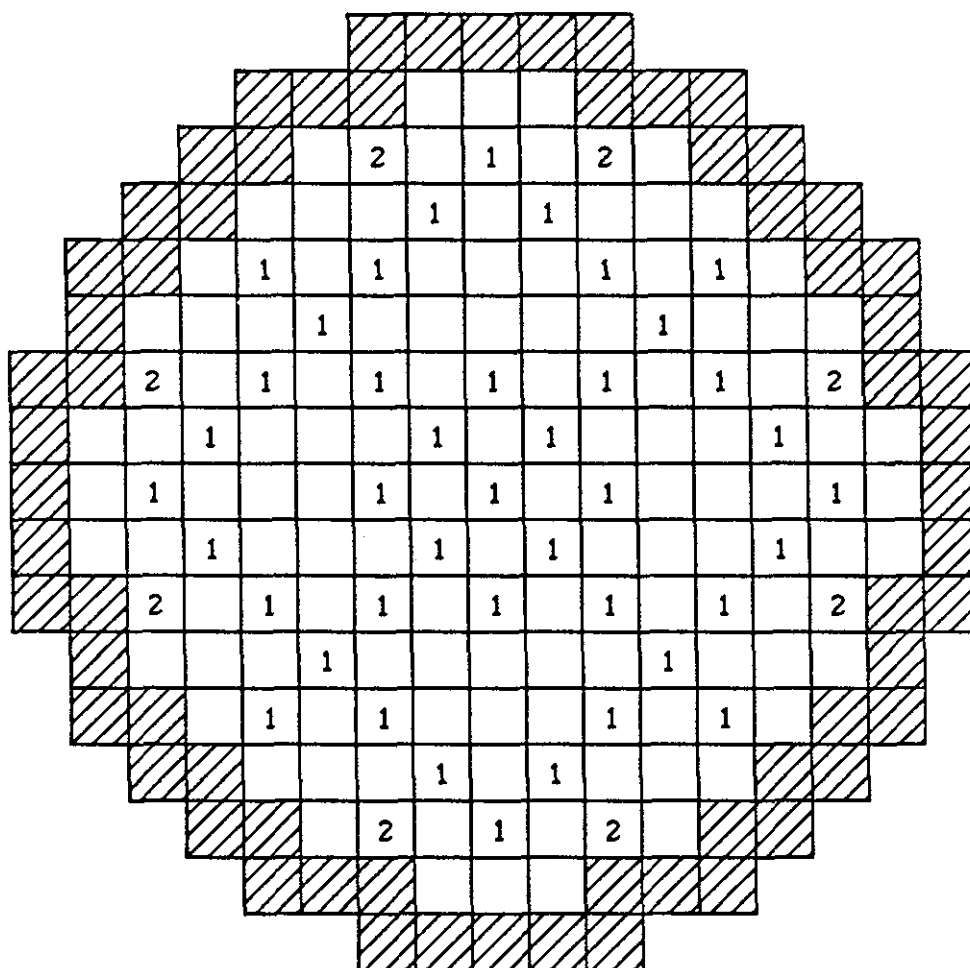


Fig.2.1 - Cross section of the reactor core

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1,2 = type of CA cross section increment

Fig.2.2 - Arrangement of Control Assemblies

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Pressurized Water Reactor

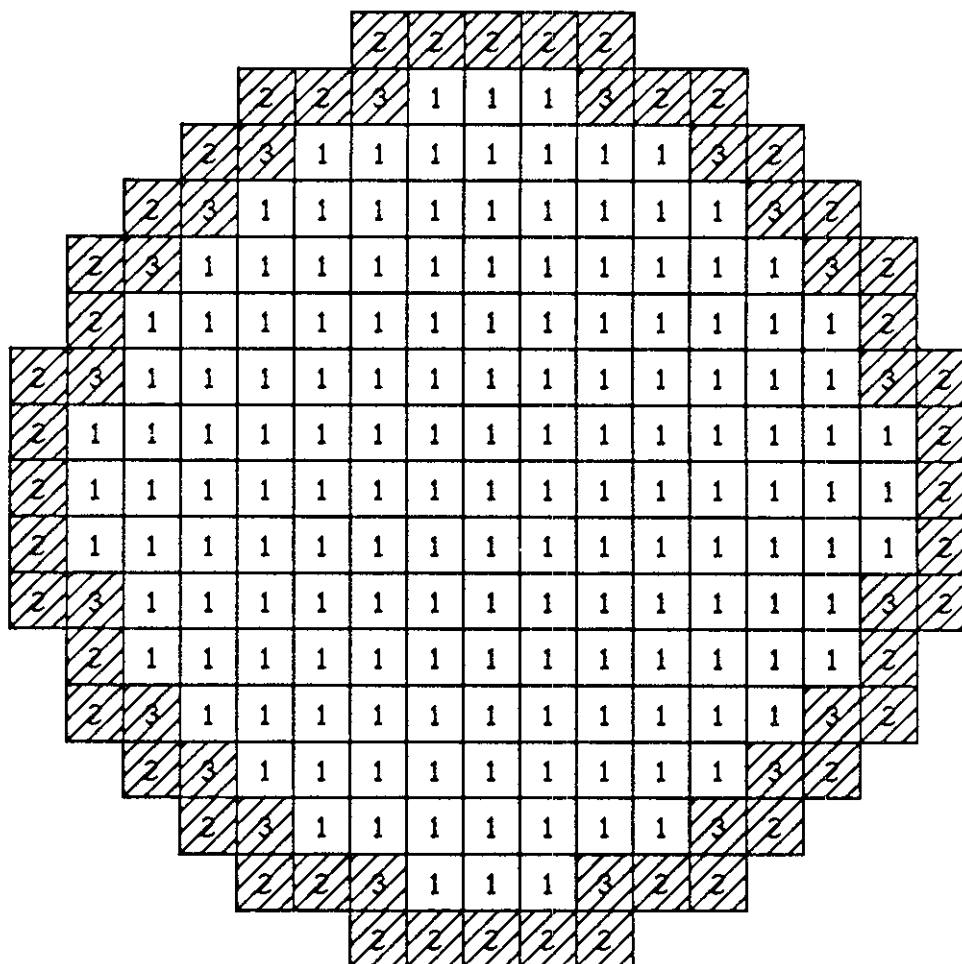


Fig.2.3 - Composition numbers in axial layers 1 and 18 (bottom and top reflector)

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Pressurized Water Reactor

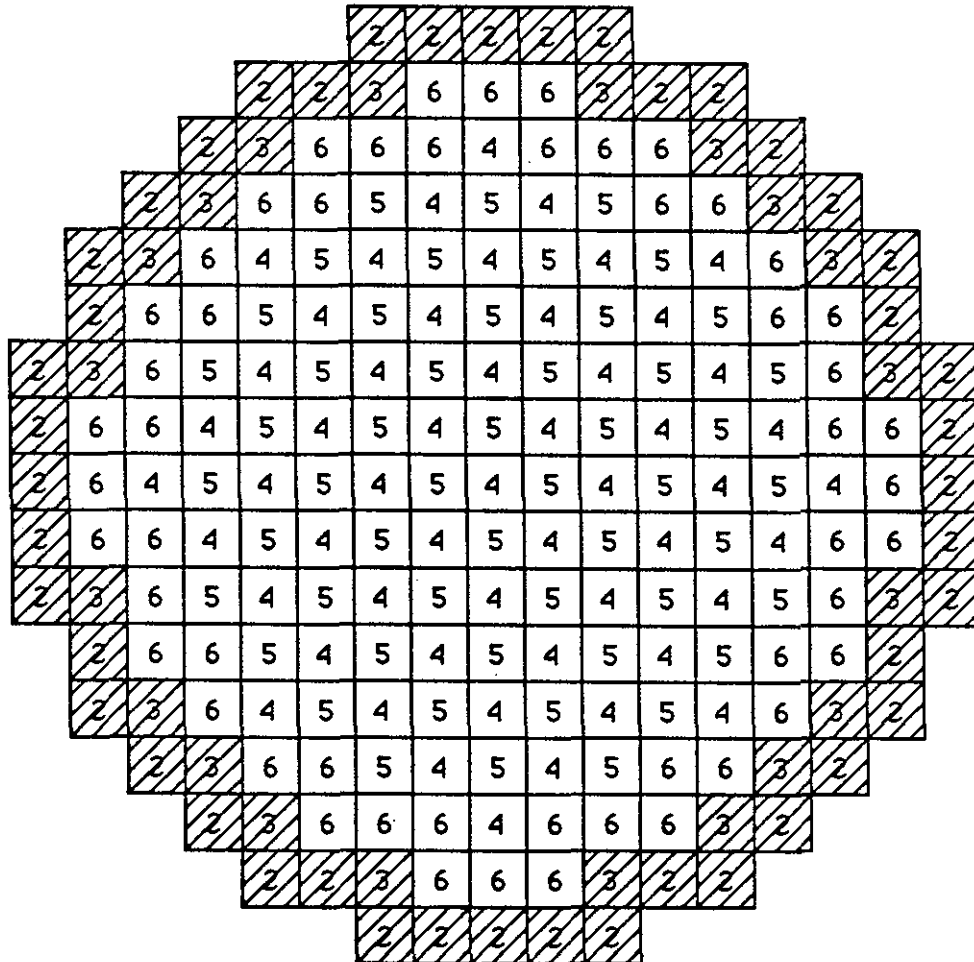


Fig.2.4 - Composition numbers in axial layer 2 (bottom layer of active core)

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Pressurized Water Reactor

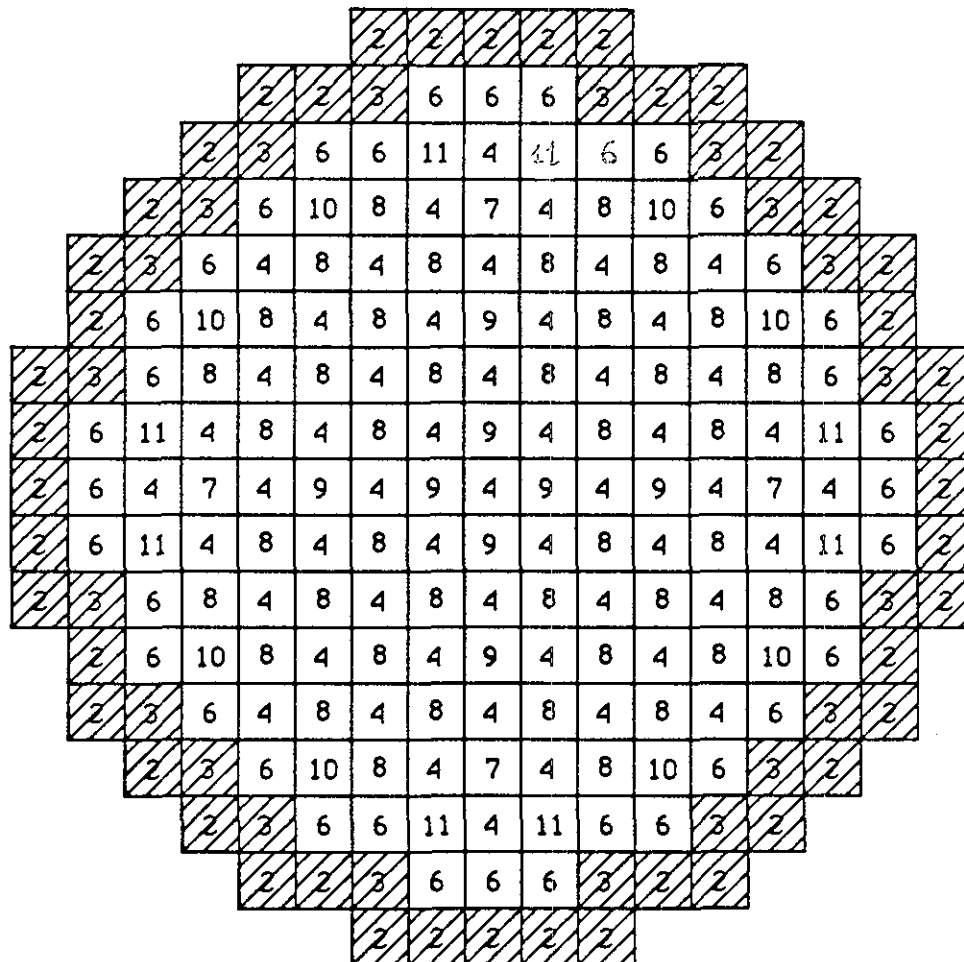
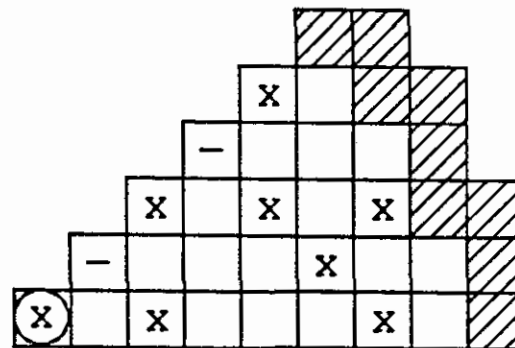


Fig.2.5 - Composition numbers in axial layers 3 through 17 (active core)

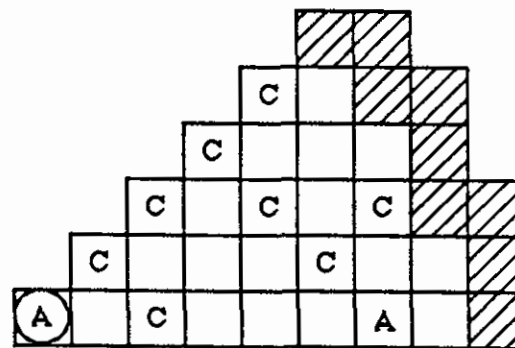
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(X) = CA to be ejected

CA type	X	—
Position in steps	0	228

Fig.3.1 - Case A1: Initial configuration of control assemblies

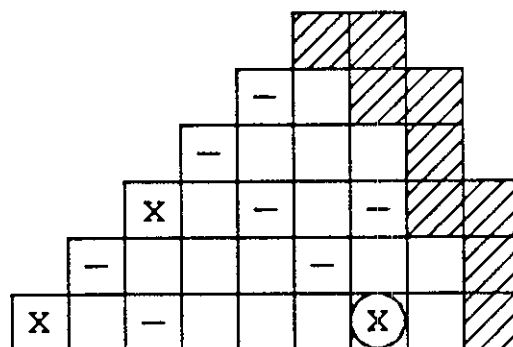


(A) = CA to be ejected

CA type	A	C
Position in steps	100	200

Fig.3.2 - Case A2: Initial configuration of control assemblies

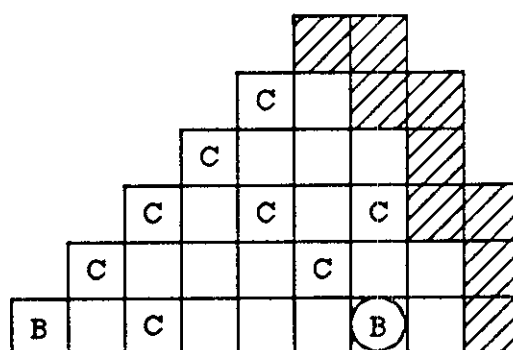
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(X) = CA to be ejected

CA type	X	—
Position in steps	0	228

Fig.3.3 - Case B1: Initial configuration of control assemblies

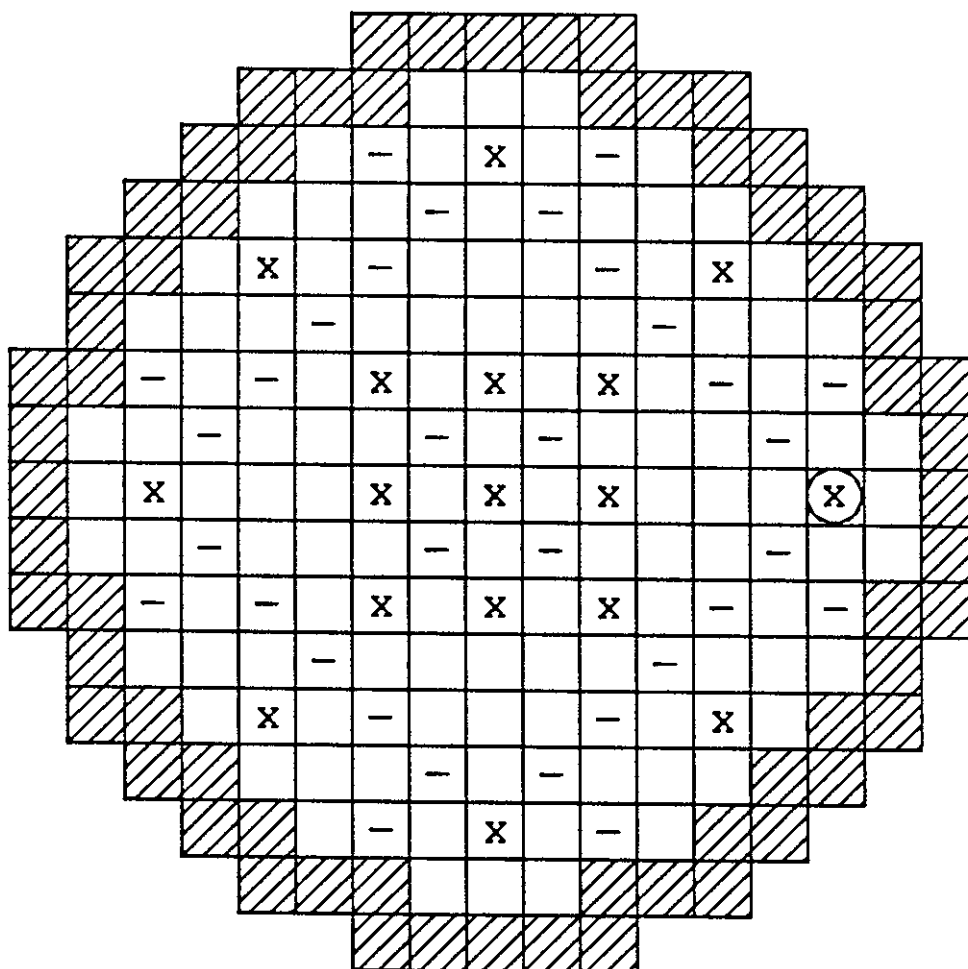


(B) = CA to be ejected

CA type	B	C
Position in steps	150	200

Fig.3.4 - Case B2: Initial configuration of control assemblies

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Pressurized Water Reactor

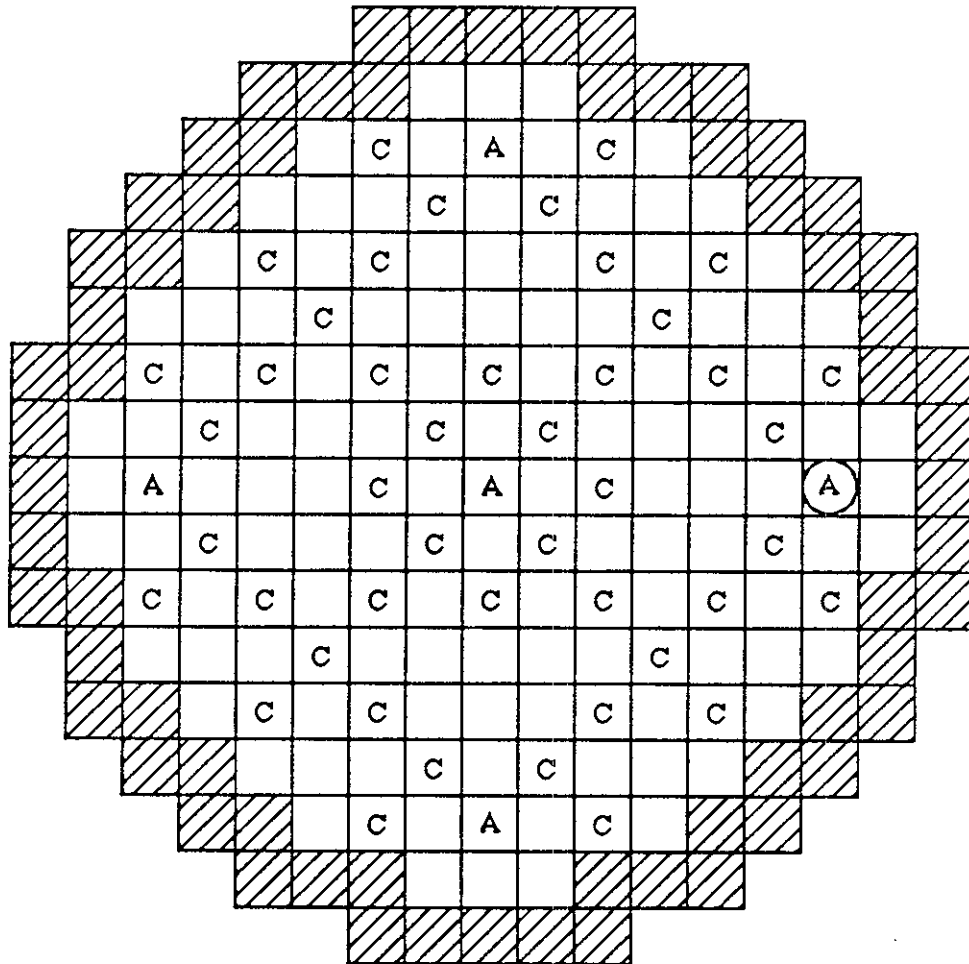


(X) = CA to be ejected

CA type	X	-
Position in steps	0	228

Fig.3.5 - Case C1: Initial configuration of control assemblies

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(A) = CA to be ejected

CA type	A	C
Position in steps	100	200

Fig. 3.6 - Case C2: Initial configuration of control assemblies

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Boiling Water Reactor

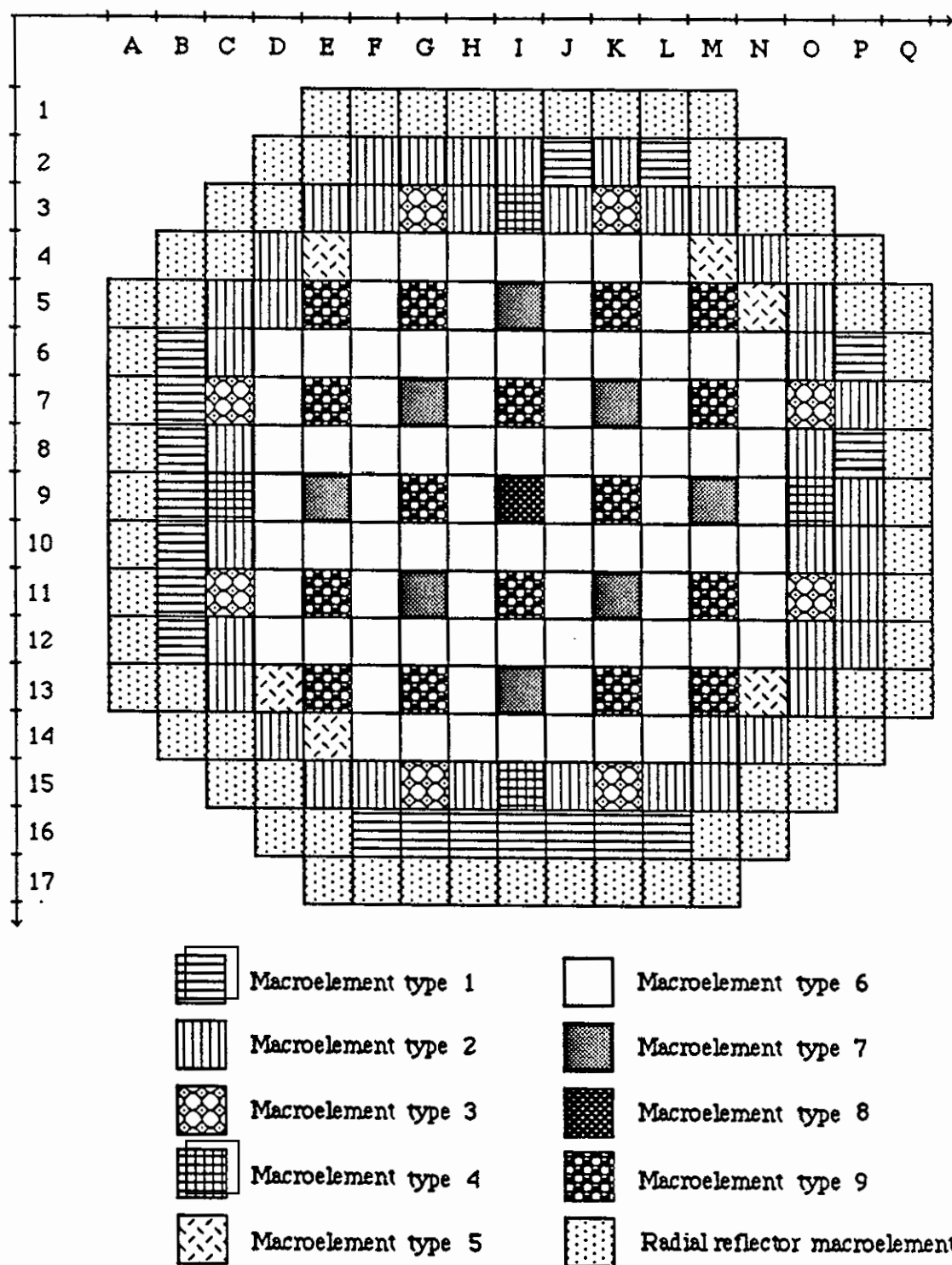


Fig.5.1 - BWR initial map

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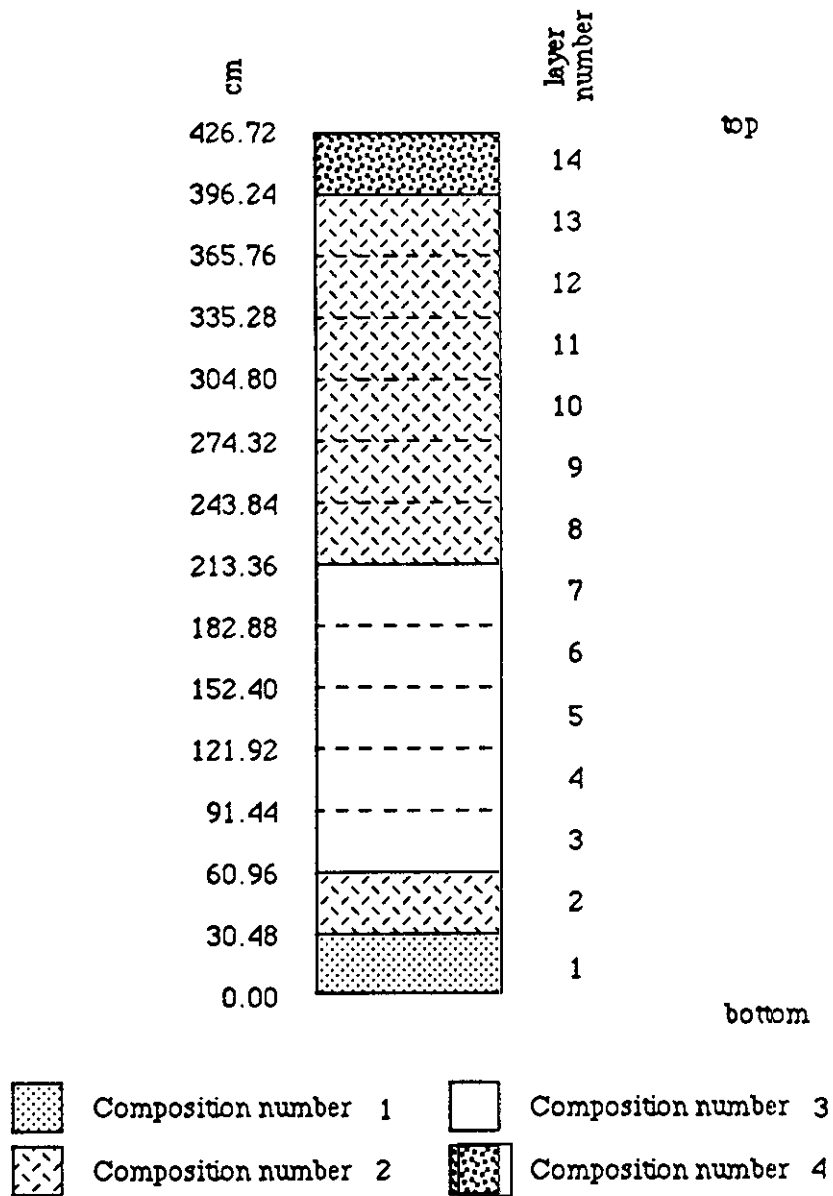


Fig. 5.2 - BWR macroelement type 1

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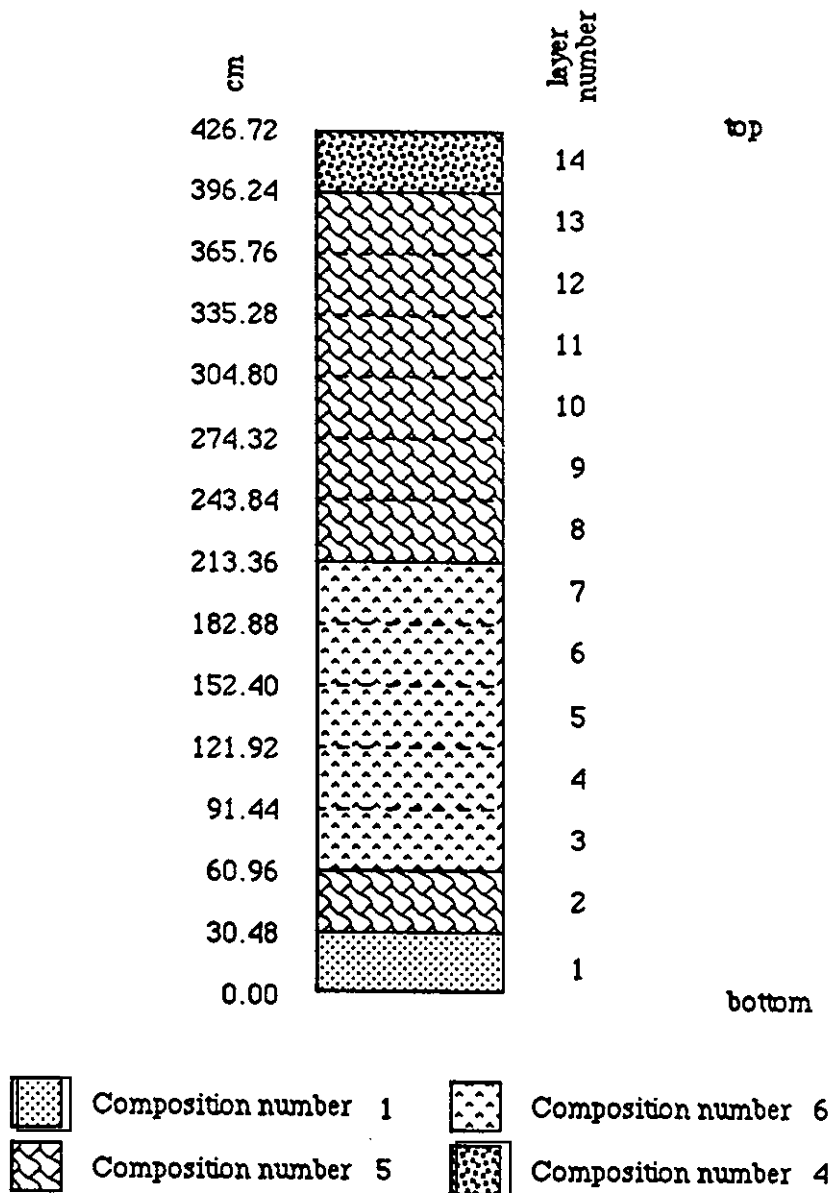


Fig.5.3 - BWR macroelement type 2

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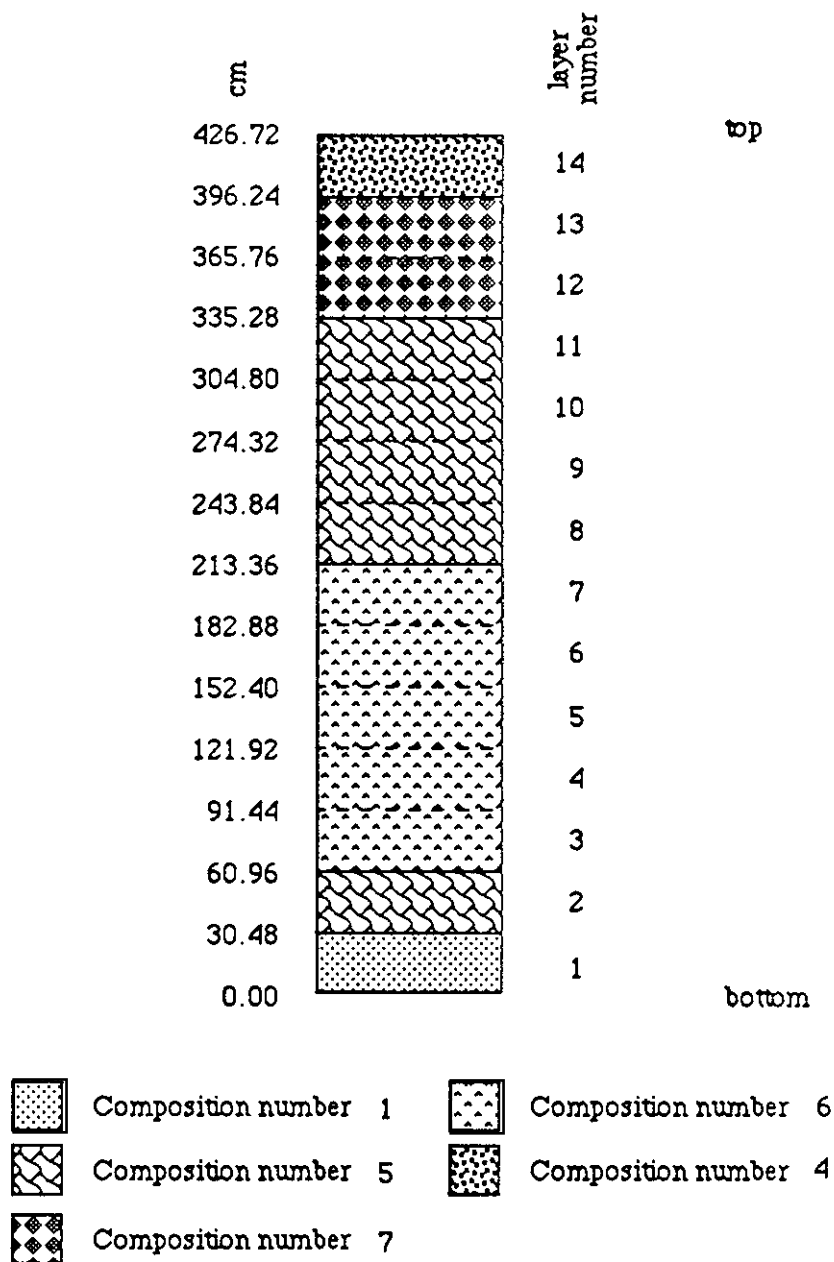


Fig 5.4 - BWR macroelement type 3

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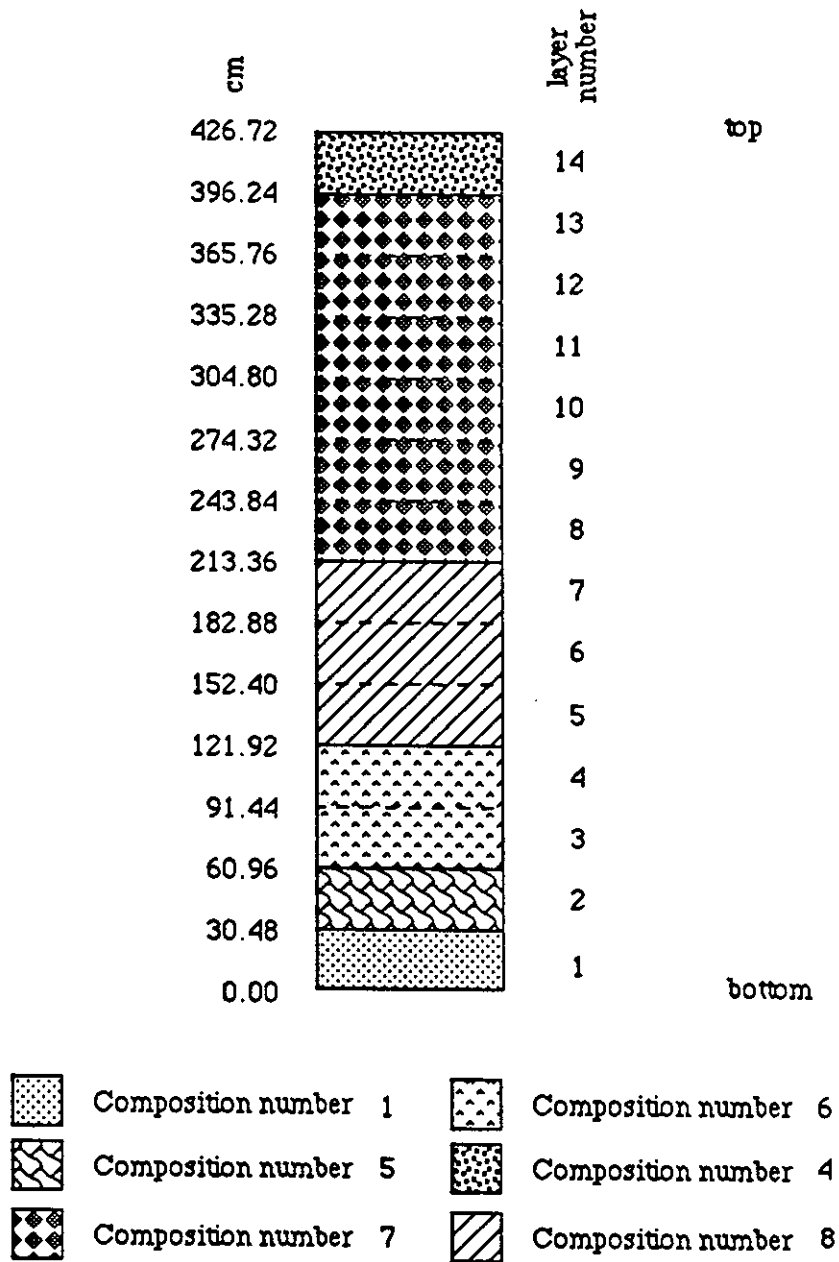


Fig.5.5 - BWR macroelement type 4

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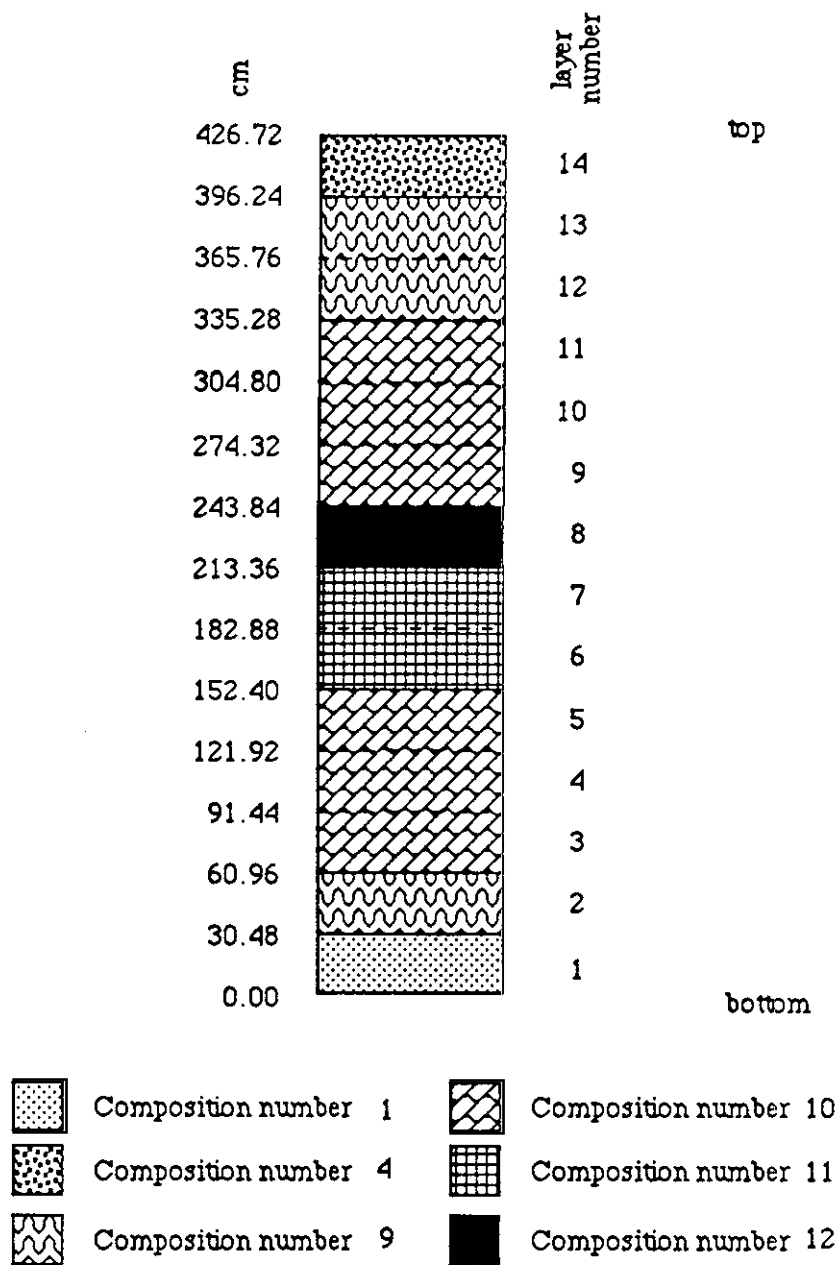


Fig.5.6 - BWR macroelement type 5

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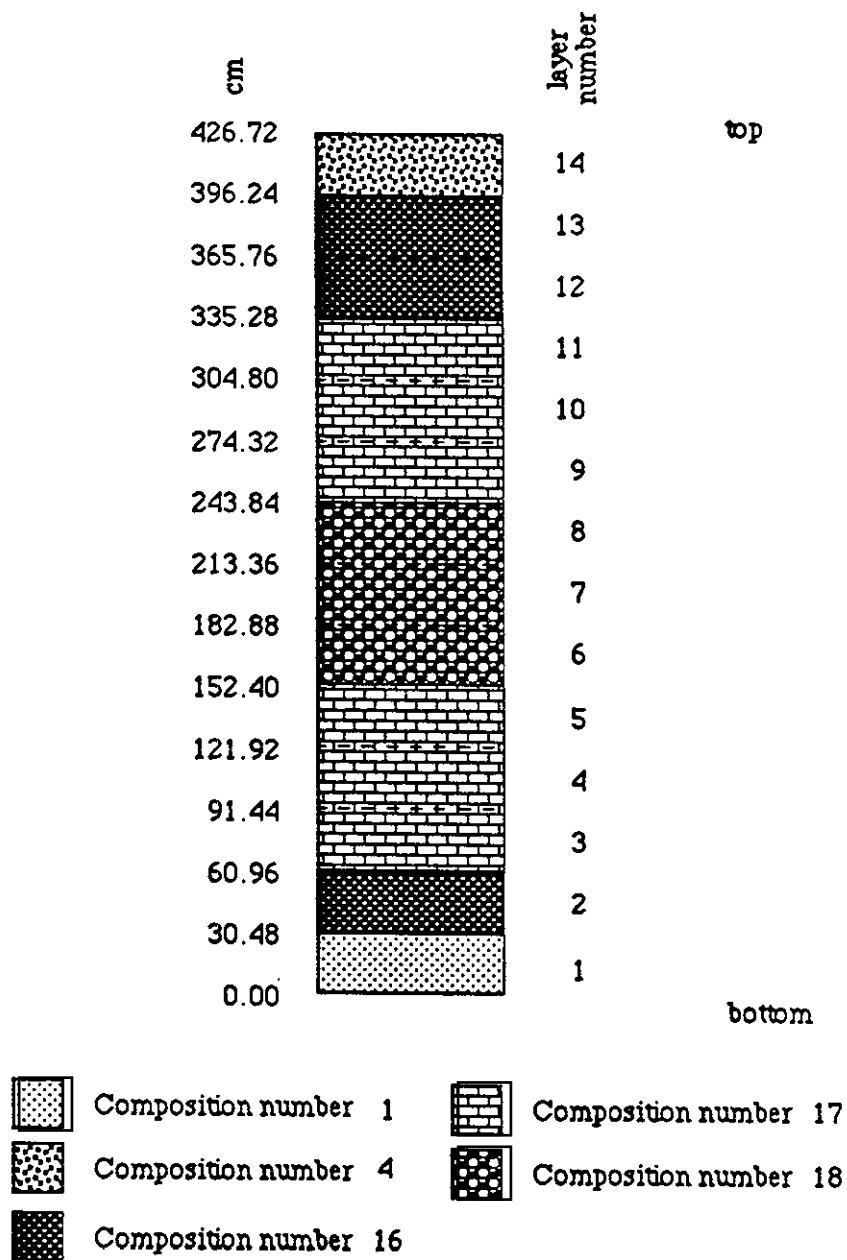


Fig.5.7 - BWR macroelement type 6

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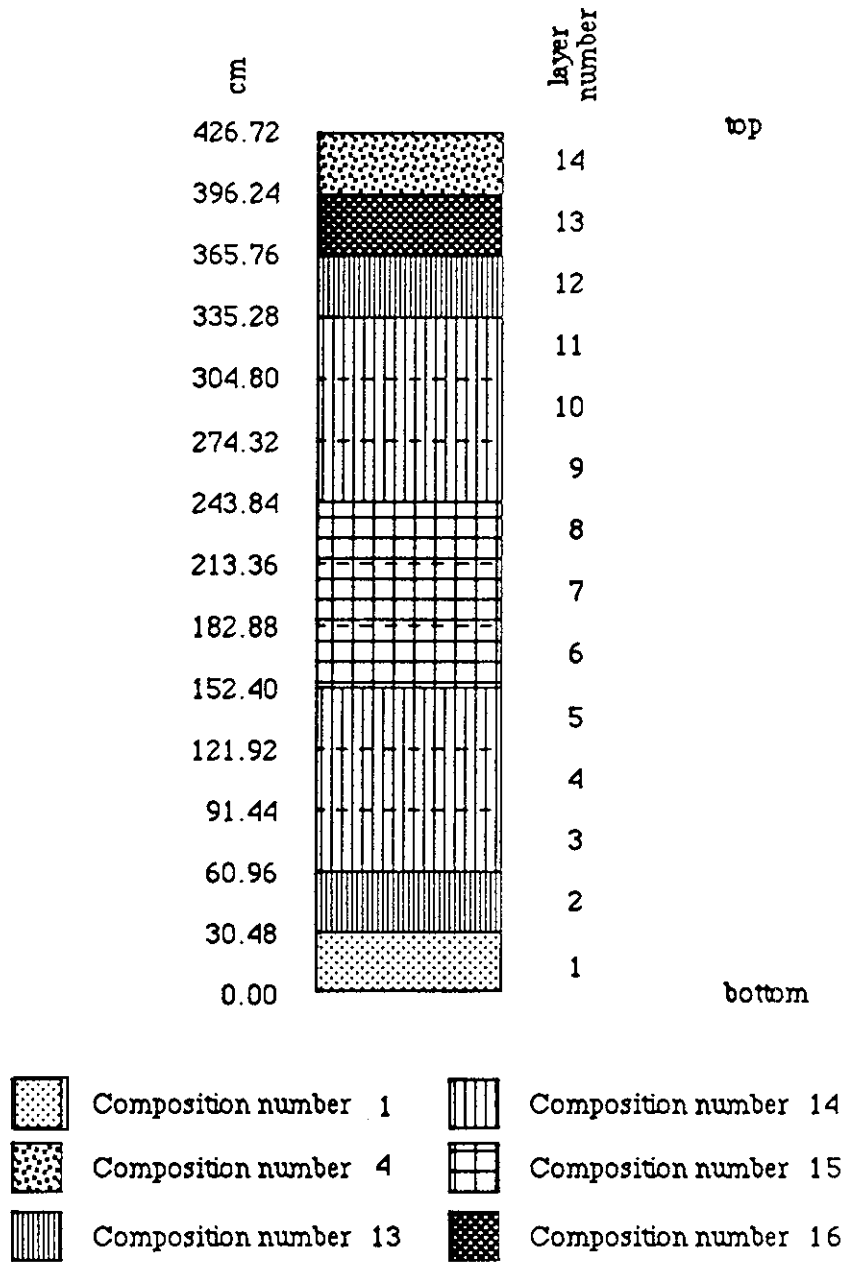


Fig.5.8 - BWR macroelement type 7

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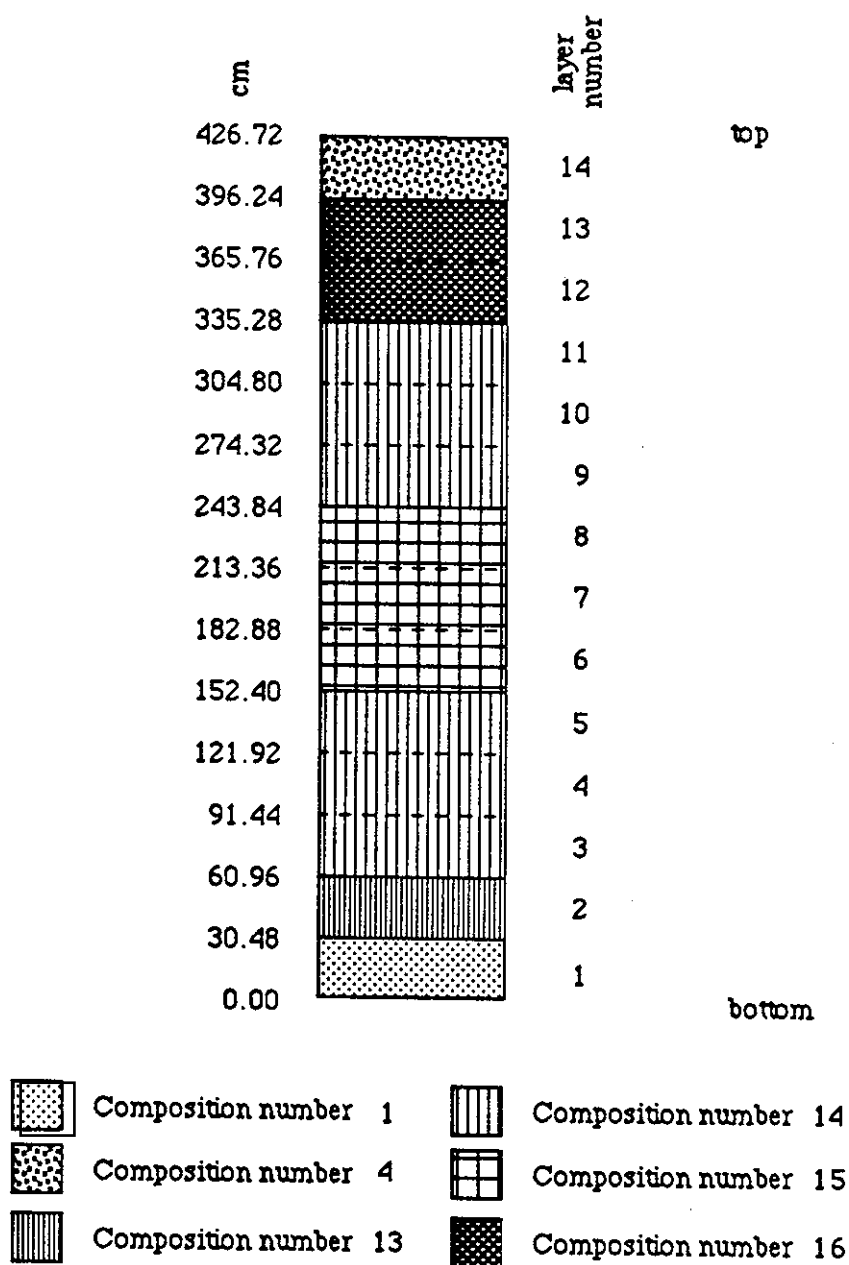


Fig.5.9- BWR macroelement type 8

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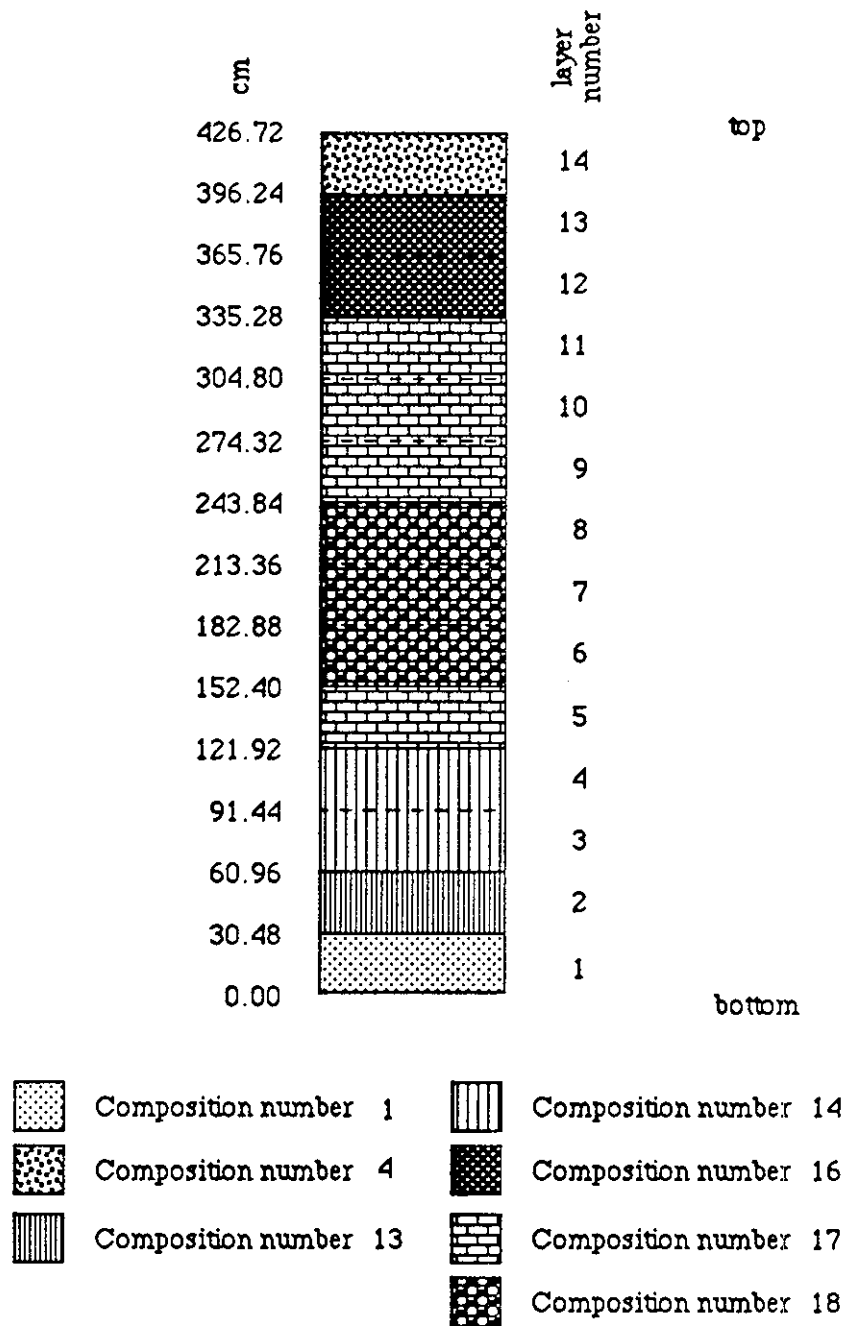
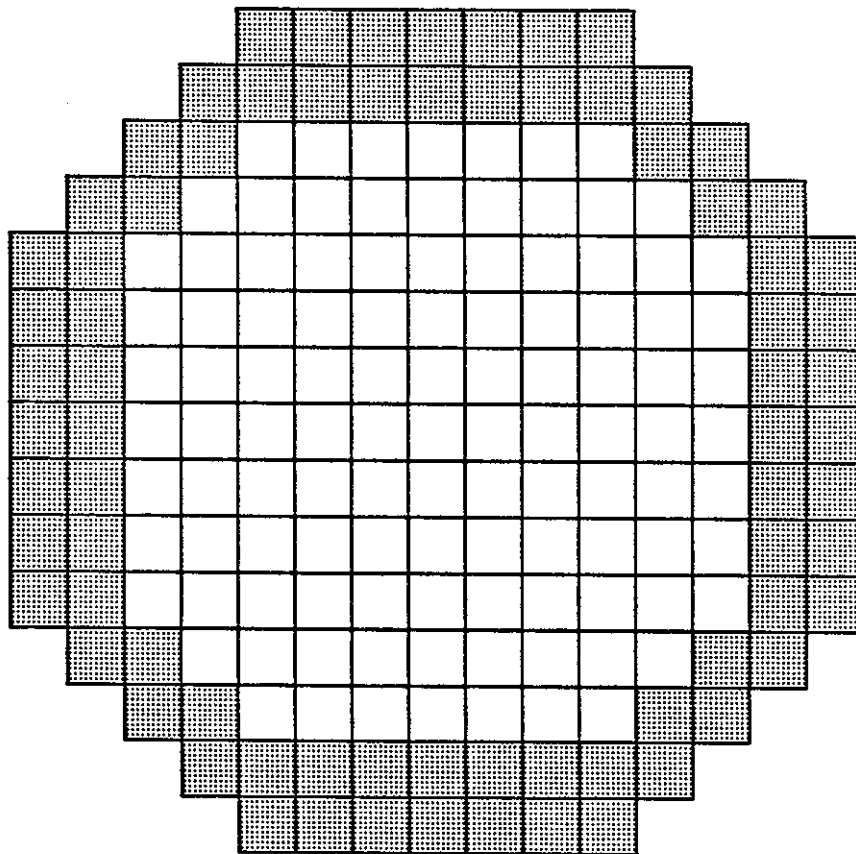


Fig.5.10 - BWR macroelement type 9

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-  macroelement with standard inlet orifice (see Figure 5.12)
-  macroelement with reduced inlet orifice (see Figure 5.13)

Fig.5.11 - Map of the macroelement inlet orifices

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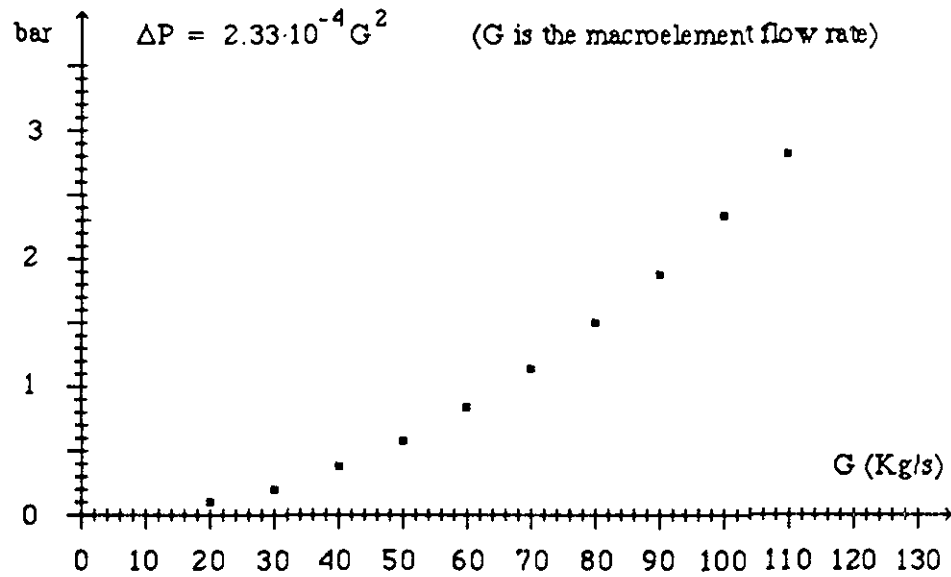


Fig. 5.12 - Inlet pressure drop vs. flow rate (standard macroelement)

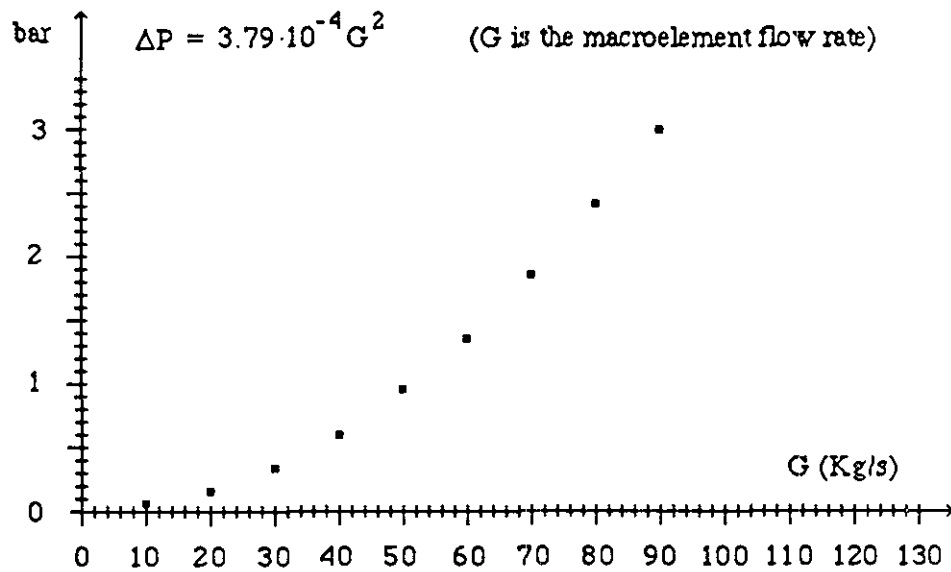


Fig. 5.13 - Inlet pressure drop vs. flow rate (peripheral macroelement)

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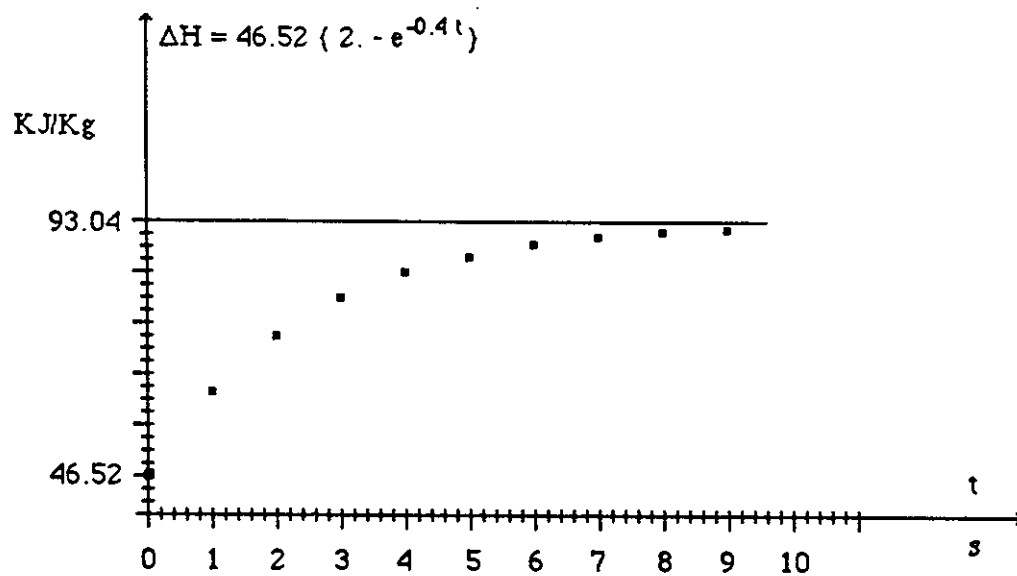


Fig.6.1. - CASE D: inlet subcooling vs. time

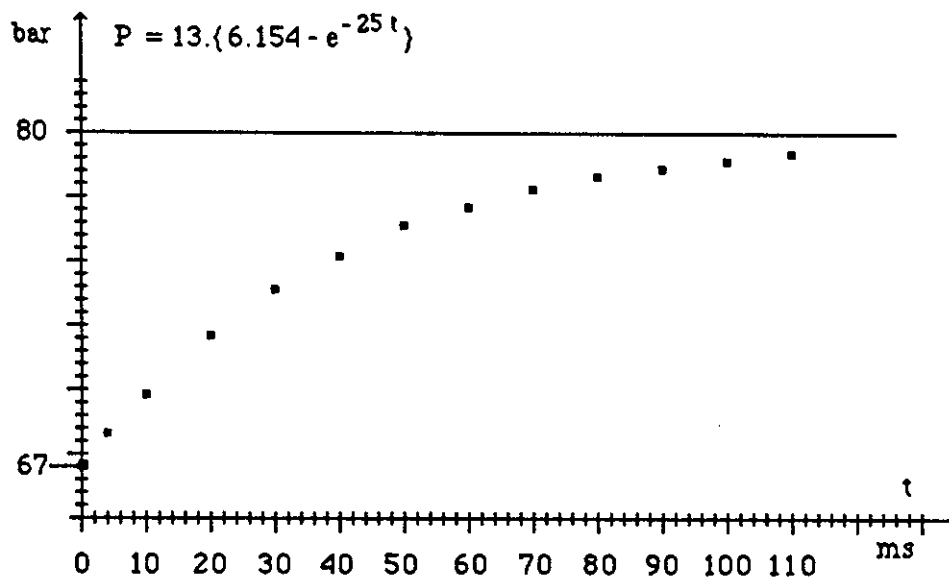


Fig.6.2. - CASE E: core pressure vs. time

[illegible]

A pyramid structure made of blocks. The bottom row consists of 8 blocks. The second row from the bottom has 7 blocks, starting from the left edge of the bottom row. The third row has 6 blocks, also starting from the left edge. The top row has 2 blocks, centered on the 4th and 5th blocks of the row below it.

99 72

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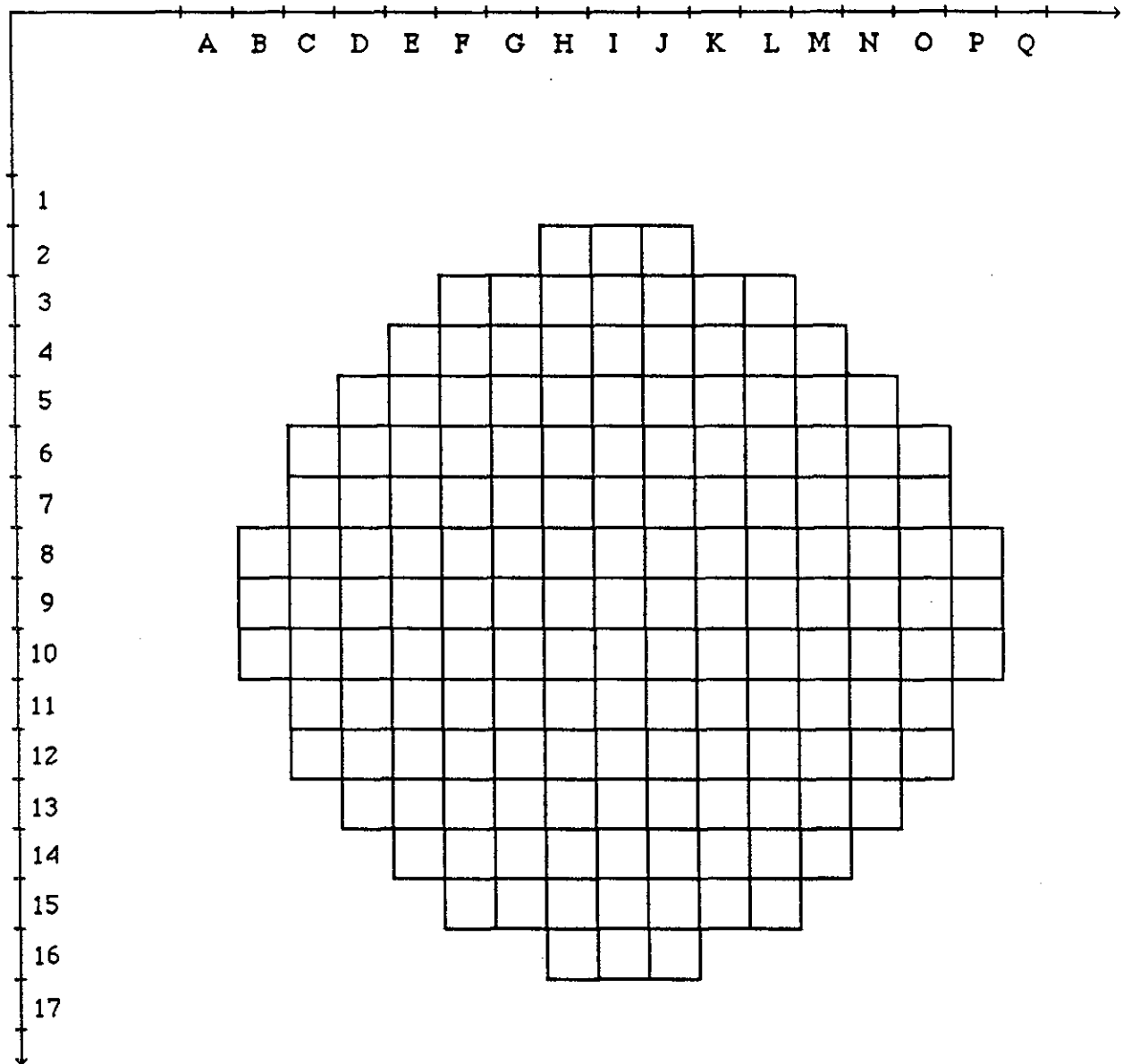


Fig.8.3 - Form for power radial distribution in Cases C1-C2

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layers of the active core

bottom												top
	1	2	3	4	5	6	7	8	9	10	11	12

Fig. 8.4 - Form for axial distributions

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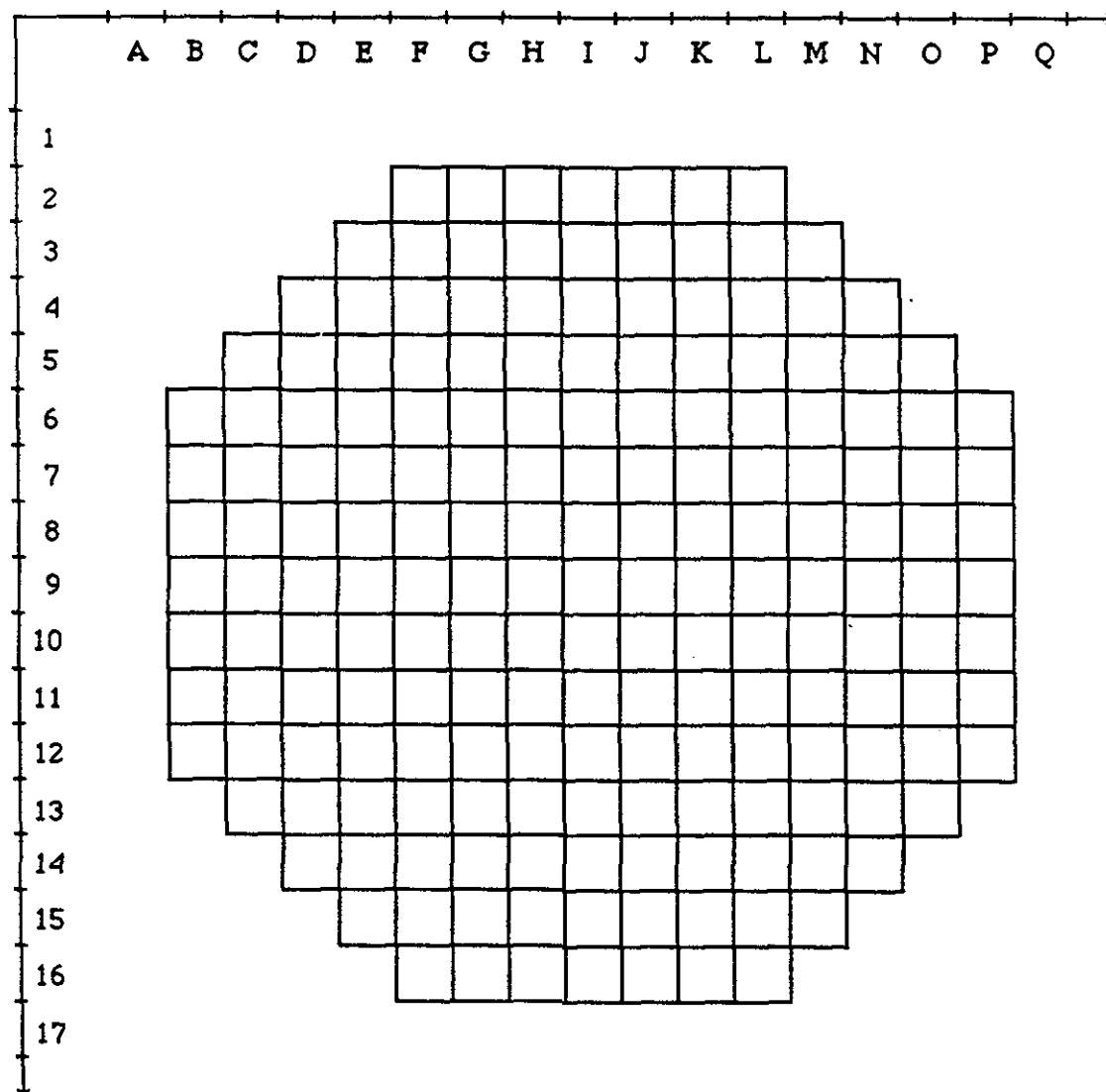


Fig.8.5 - Form for radial distributions in Cases D1-E1